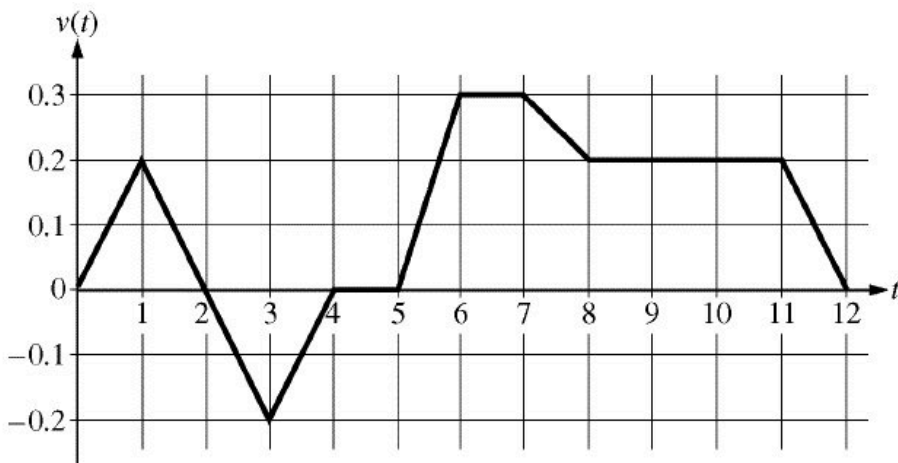


2-3-3

1.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t=0$ minutes and arriving at school at time $t=12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

Find the acceleration of Caren's bicycle at time $t=7.5$ minutes. Indicate units of measure.

Part A

1 point is earned for answer

1 point is earned for unit

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$



0	1	2
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The student response earns two of the following points:

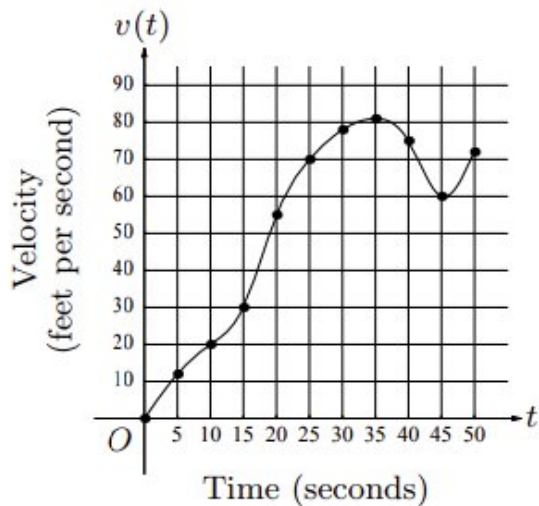
1 point is earned for answer

1 point is earned for unit

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

2-3-3

2.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph

Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.

Part C

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

2-3-3



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

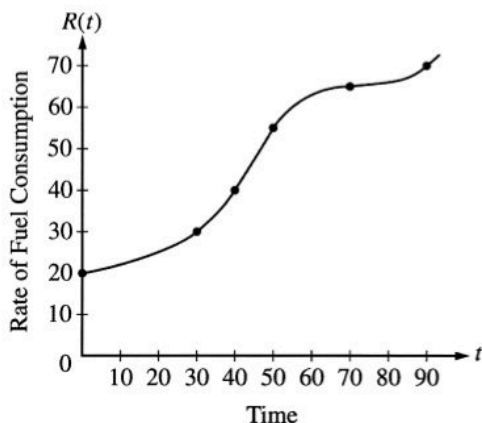
$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

2-3-3

3.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min² with correct units



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min² with correct units

2-3-3

4.



t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

2-3-3 $= 1.017$ (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the value

1 point is earned for the interpretation with units

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

2-3-3



0	1	2	3
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The student earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for approximation

1 point is earned for underestimating with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$

2-3-3

5.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to estimate $w'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

Part A

One point is earned for estimate

One point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
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The student earns all of the following points:

One point is earned for estimate

One point is earned for interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

2-3-3

6.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

(a) Use the data in the table to approximate Ben's acceleration at time $t = 5$ seconds. Indicate units of measure.

(b) Using correct units, interpret the meaning of $\int_0^{60} |v(t)| dt$ in the context of this problem. Approximate

$\int_0^{60} |v(t)| dt$ using a left Riemann sum with the subintervals indicated by the data in the table.

(c) For $40 \leq t \leq 60$, must there be a time t when Ben's velocity is 2 meters per second? Justify your answer.

(d) A light is directly above the western end of the track. Ben rides so that at time t , the distance $L(t)$ between Ben and the light satisfies $(L(t))^2 = 12^2 + (B(t))^2$. At what rate is the distance between Ben and the light changing at time $t = 40$?

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

1 point is earned for units in (a) or (b)

Part B

1 point is earned for meaning of the integral

2-3-3

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139$ meters

1 point is earned for units in (a) or (b)



0	1	2	3
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The student response earns all of the following points:

1 point is earned for meaning of the integral

1 point is earned for the approximation

$\int_0^{60} |v(t)| dt$ is the total distance, in meters, that Ben rides over the 60-second interval $t = 0$ to $t = 60$.

$\int_0^{60} |v(t)| dt \approx 2.0 \cdot 10 + 2.3(40 - 10) + 2.5(60 - 40) = 139$ meters

1 point is earned for units in (a) or (b)

Part C

1 point is earned for difference quotient

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for difference quotient

2-3-3

1 point is earned for conclusion with justification

Because $\frac{B(60) - B(40)}{60 - 40} = \frac{49 - 9}{20} = 2$, the Mean Value Theorem implies there is a time t , $40 < t < 60$, such that $v(t) = 2$.

Part D

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the derivatives

1 point is earned if uses $B'(t) = v(t)$

1 point is earned for the answer

$$2L(t)L'(t) = 2B(t)B'(t)$$

$$L'(40) = \frac{B(40)v(40)}{L(40)} = \frac{9 \cdot 2.5}{\sqrt{144+81}} = \frac{3}{2} \text{ meters/sec}$$

2-3-3

7.

t (seconds)	0	10	40	60
$B(t)$ (meters)	100	136	9	49
$v(t)$ (meters per second)	2.0	2.3	2.5	4.6

Ben rides a unicycle back and forth along a straight east-west track. The twice-differentiable function B models Ben's position on the track, measured in meters from the western end of the track, at time t , measured in seconds from the start of the ride. The table above gives values for $B(t)$ and Ben's velocity, $v(t)$, measured in meters per second, at selected times t .

Use the data in the table to approximate Ben's acceleration at time $t=5$ seconds. Indicate units of measure.

Part A

1 point is earned for answer

$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for answer

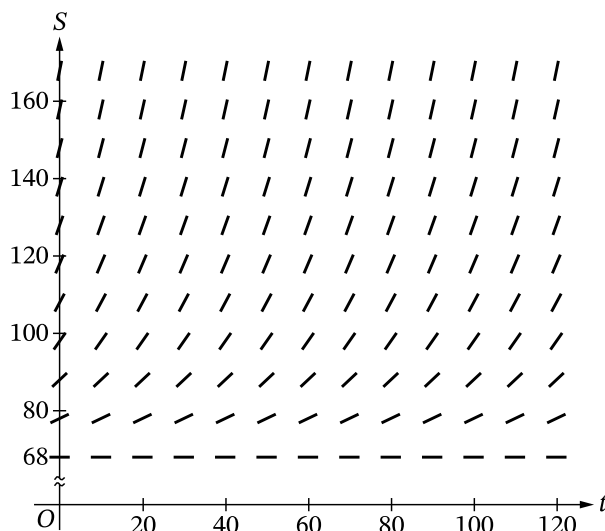
$$a(5) \approx \frac{v(10) - v(0)}{10 - 0} = \frac{0.3}{10} = 0.03 \text{ meters/sec}^2$$

2-3-3

8. For $0 \leq t \leq 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^{\circ}\text{F}$) and t is measured in minutes. The temperature of the bowl of soup is always greater than 68°F and is 148°F at time $t = 0$.

Part A

Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.

**Part B**

Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

Part C

According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

Part D

Using separation of variables, find the particular solution $y = S(t)$ to the differential equation

$$\frac{dS}{dt} = -\frac{1}{20}(S - 68) \text{ with initial condition } S(0) = 148.$$

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** can be earned for a response of “because there are no negative slopes” or “because the slopes are positive.”
- **P1** can be earned by using a local argument at any point (t, S) with $0 \leq t \leq 120$ and $S > 68$, rather than a global argument.

2-3-3

Model Solution:

The slopes of the line segments in a slope field for the given differential equation should all be negative because for $S > 68$, $\frac{dS}{dt} < 0$. However, the slopes of the line segments in the slope field given are all nonnegative.



0	1
---	---

The response accurately includes the criteria below.

- Explanation

Part B Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response can earn **P2** with $\frac{dS}{dt} \big|_{(0, 148)} = -4$, “slope is -4 ,” or equivalent.
- A response that presents a linear approximation with a slope of -4 also earns **P2**. For example, a response of $148 - 4(5)$ earns **P2**.
- A response that declares the slope equal to any nonzero value $k \neq -4$ does not earn **P2**.

Model Solution:

$$\frac{dS}{dt} \big|_{(0, 148)} = -\frac{1}{20}(148 - 68) = -4$$



0	1
---	---

The response accurately includes the criteria below.

- Slope of tangent line

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response that declares the slope equal to any nonzero value $k \neq -4$:
 - Earns **P3** if there is evidence that the given differential equation was used to calculate k and that k was used correctly in a tangent line approximation of the form $148 + k(5)$. Subsequent errors in simplification will not be considered in scoring for **P3**.
 - Does not earn **P3** if k is simply declared to be a slope without a connection to the differential equation.
- **P3** cannot be earned for an approximation at any value of t other than 5.

2-3-3

- A response does not have to present the tangent line equation but must clearly demonstrate its use at $t = 5$ in finding the requested approximation in order to earn **P3**.
- A response of $148 - 4(5)$ earns **P3**. Subsequent errors in simplification will not be considered in scoring for **P3**.

Model Solution:

$$S(5) \approx 148 - 4(5) = 128$$

The temperature of the soup at $t = 5$ minutes is approximately 128°F .



0	1
---	---

The response accurately includes the criteria below.

- Tangent line approximation

Part C Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P4**, a response must include one of the following: $\frac{d^2S}{dt^2}$, $-\frac{1}{20}\frac{dS}{dt}$, $\frac{1}{400}(S - 68)$, “the second derivative,” or equivalent.
 - “ S' is increasing (or decreasing)” or a reference to concavity without a connection to the second derivative is not sufficient to earn **P4**.

Model Solution:

$$\text{Because } S(t) > 68, \frac{d^2S}{dt^2} = -\frac{1}{20}\frac{dS}{dt} = \left(-\frac{1}{20}\right)\left(-\frac{1}{20}\right)(S - 68) = \frac{1}{400}(S - 68) > 0.$$



0	1
---	---

The response accurately includes the criteria below.

- Considers $\frac{d^2S}{dt^2}$

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To be eligible to earn **P5**, a response must have earned **P4**.

2-3-3

- Responses do not need to say “concave up” to earn **P5**. A response of “underestimate because the second derivative is positive” is sufficient to earn **P5**.

Model Solution:

Because the second derivative is always positive, the tangent line approximation is an underestimate of the temperature predicted by the model.



0	1
---	---

The response accurately includes the criteria below.

- Answer with reason

Part D Point 6

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P6**.
- A response earns **P6** if the separation of variables results in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants.
- Special case: If the response begins with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P6**.

Model Solution:

$$\frac{dS}{S-68} = -\frac{1}{20}dt$$



0	1
---	---

The response accurately includes the criteria below.

- Separate variables

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P7**.

2-3-3

- A response is eligible to earn **P7** if the separation of variables resulted in an equation of the form $A \frac{dS}{S-68} = B dt$, where A and B are nonzero constants. A response that presents consistent antiderivatives of the form $A \ln(S - 68)$ (with either parentheses or absolute value) and Bt earns **P7**.
- Special case: If the response began with a separation of the form $A \frac{dS}{S+68} = B dt$, where A and B are correct coefficients, it earns **P7** with two consistent antiderivatives.
- A response with no constant of integration is eligible to earn **P7**.

Model Solution:

$$\ln|S - 68| = -\frac{t}{20} + C$$



0	1
---	---

The response accurately includes the criteria below.

- Finds antiderivatives

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response with no separation of variables does not earn **P8**.
- A response with no constant of integration does not earn **P8**.
- A response is eligible for **P8** if it has earned P6 and has one of the two antiderivatives correct OR if it has earned **P7**.
- An eligible response earns **P8** by correctly including the constant of integration in an equation and substituting 0 for t and 148 for S .

Model Solution:

$$\ln|148 - 68| = -\frac{0}{20} + C \Rightarrow \ln 80 = C$$

$$\ln(S - 68) = -\frac{t}{20} + \ln 80$$



0	1
---	---

The response accurately includes the criteria below.

- Constant of integration and uses initial condition

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

2-3-3

Scoring Notes:

- A response is eligible for **P9** only if it has earned **P6**, **P7**, and **P8**.
- A response earns **P9** only for an answer of $S(t) = 68 + 80e^{-t/20}$ or equivalent.

Model Solution:

$$S - 68 = e^{-t/20 + \ln 80} = 80e^{-t/20}$$

$$S(t) = 68 + 80e^{-t/20}$$



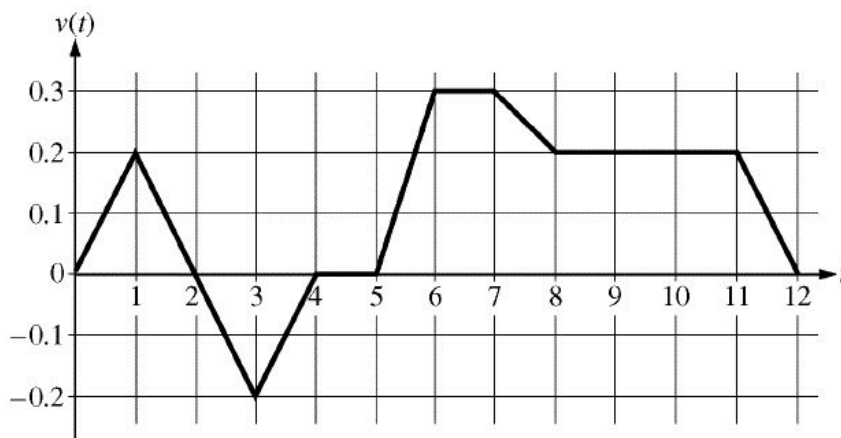
0	1
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The response accurately includes the criteria below.

- Solves for $S(t)$

2-3-3

9.



Caren rides her bicycle along a straight road from home to school, starting at home at time $t = 0$ minutes and arriving at school at time $t = 12$ minutes. During the time interval $0 \leq t \leq 12$ minutes, her velocity $v(t)$, in miles per minute, is modeled by the piecewise-linear function whose graph is shown above.

(a) Find the acceleration of Caren's bicycle at time $t = 7.5$ minutes. Indicate units of measure.

(b) Using correct units, explain the meaning of $\frac{1}{12} \int_0^{12} |v(t)| dt$ in terms of Caren's trip. Find the value of $\frac{1}{12} \int_0^{12} |v(t)| dt$.

(c) Shortly after leaving home, Caren realizes she left her calculus homework at home, and she returns to get it. At what time does she turn around to go back home? Give a reason for your answer.

(d) Larry also rides his bicycle along a straight road from home to school in 12 minutes. His velocity is modeled by the function w given by $w(t) = \frac{\pi}{15} \sin\left(\frac{\pi}{12}t\right)$, where $w(t)$ is in miles per minute for $0 \leq t \leq 12$ minutes. Who lives closer to school: Caren or Larry? Show the work that leads to your answer.

Part A

1 point is earned for the answer

1 point is earned for units

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles / minute}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for units

2-3-3

$$a(7.5) = v'(7.5) = \frac{v(8) - v(7)}{8 - 7} = -0.1 \text{ miles /minute}^2$$

Part B

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the meaning of integral

1 point is earned for the value of integral

$\int_0^{12} |v(t)| dt$ is the total distance, in miles, that Caren rode during the 12 minutes from $t = 0$ to $t = 12$.

$$\int_0^{12} |v(t)| dt = \int_0^2 v(t) dt - \int_2^4 v(t) dt + \int_4^{12} v(t) dt$$

$$= 0.2 + 0.2 + 1.4 = 1.8 \text{ miles}$$

Part C

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative.

2-3-3



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

1 point is earned for reason

Caren turns around to go back home at time $t = 2$ minutes. This is the time at which her velocity changes from positive to negative..

Part D

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$$\int_0^{12} w(t) dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

$$\int_0^{12} v(t) dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.



0	1	2	3
---	---	---	---

The students response earns all of the following points:

2 points are earned for Larry's distance from school where 1 point is earned for integral and 1 point is earned for value

1 point is earned for Caren's distance from school and conclusion

$$\int_0^{12} w(t) dt = 1.6; \text{ Larry lives 1.6 miles from school.}$$

$$\int_0^{12} v(t) dt = 1.4; \text{ Caren lives 1.4 miles from school.}$$

Therefore, Caren lives closer to school.

2-3-3

10.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

Estimate $f'(4)$. Show the work that leads to your answer.

Part A

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$



0	1
---	---

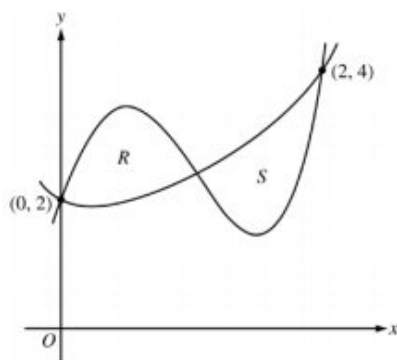
The student response earns one of the following points:

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$

2-3-3

11.



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.

 Let h be the vertical distance between the graphs of f and g in region S . Find the rate at which h changes with respect to x when $x = 1.8$.

Part C

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$

2-3-3

12. Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.



Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.

Part B

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$A'(15) = -0.164$ (or -0.163) The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interpretation

1 point is earned for $A'(15)$

$A'(15) = -0.164$ (or -0.163)

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.

13. The graph of $y = e^{\tan x} - 2$ crosses the x -axis at one point in the interval $[0, 1]$. What is the slope of the graph at this point?

(A) 0.606

(B) 2

(C) 2.242

(D) 2.961

(E) 3.747



2-3-3

14. On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos(t^2/18)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.



(a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587) The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.



0	1	2
---	---	---

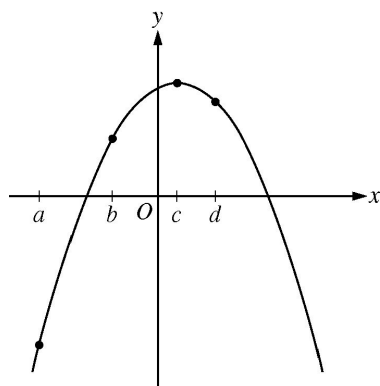
The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$G'(5) = -24.588$ (or -24.587) The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

15.



Graph of f

The figure above shows the graph of a function f . Which of the following has the greatest value?

(A) $f(a)$

(B) $f'(a)$

(C) $f'(c)$

(D) $f(d) - f(c)$

(E) $\frac{f(b) - f(a)}{b - a}$



2-3-3

16.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t=0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t=0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

Part A

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.0170 F per minute at time

$t=12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6}$$

$$= 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017 oF per minute at time $t=12$ minutes.

2-3-3

17.

x	2.5	2.8	3.0	3.1
$f(x)$	31.25	39.20	45	48.05

The function f is differentiable and has values as shown in the table above. Both f and f' are strictly increasing on the interval $0 \leq x \leq 5$. Which of the following could be the value of $f'(3)$?

- (A) 20
 (B) 27.5
 (C) 29
 (D) 30
 (E) 30.5



t (minutes)	0	4	8	12	16
$H(t)$ ($^{\circ}\text{C}$)	65	68	73	80	91

The temperature, in degrees Celsius ($^{\circ}\text{C}$), of an oven being heated is modeled by an increasing differentiable function H of time t , where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

18. Use the data in the table to estimate the instantaneous rate at which the temperature of the oven is changing at time $t = 10$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for difference quotient

1 point is earned for answer with units

$$H'(10) = \frac{H(12) - H(8)}{12 - 8} = \frac{80 - 73}{4} = \frac{7}{4}^{\circ}\text{C}/\text{min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for difference quotient

1 point is earned for answer with units

$$H'(10) = \frac{H(12) - H(8)}{12 - 8} = \frac{80 - 73}{4} = \frac{7}{4}^{\circ}\text{C}/\text{min}$$

2-3-3

19. Are the data in the table consistent with or do they contradict the claim that the temperature of the oven is increasing at an increasing rate? Give a reason for your answer.

Part D

1 point is earned for considers slopes of four secant lines

1 point is earned for the explanation

1 point is earned for the conclusion consistent with explanation

If a continuous function is increasing at an increasing rate, then the slopes of the secant lines of the graph of the function are increasing. The slopes of the secant lines for the four intervals in the table are $\frac{3}{4}$, $\frac{5}{4}$, $\frac{7}{4}$, and $\frac{10}{4}$, respectively.

Since the slopes are increasing, the data are consistent with the claim.

OR

By the Mean Value Theorem, the slopes are also the values of $H'(c_k)$ for some times $c_1 < c_2 < c_3 < c_4$, respectively. Since these derivative values are positive and increasing, the data are consistent with the claim.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considers slopes of four secant lines

1 point is earned for the explanation

1 point is earned for the conclusion consistent with explanation

If a continuous function is increasing at an increasing rate, then the slopes of the secant lines of the graph of the function are increasing. The slopes of the secant lines for the four intervals in the table are $\frac{3}{4}$, $\frac{5}{4}$, $\frac{7}{4}$, and $\frac{10}{4}$, respectively.

Since the slopes are increasing, the data are consistent with the claim.

OR

By the Mean Value Theorem, the slopes are also the values of $H'(c_k)$ for some times $c_1 < c_2 < c_3 < c_4$, respectively. Since these derivative values are positive and increasing, the data are consistent with the claim.

2-3-3

20.

t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

Use the data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day, or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

2-3-3

21.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

Estimate $T'(7)$. Show the work that leads to your answer. Indicate units of measure.

Part A

1 point is earned for the answer

$$\frac{T(8)-T(6)}{8-6} = \frac{55-62}{2} = -\frac{7}{2}^{\circ}\text{C}/\text{cm}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{T(8)-T(6)}{8-6} = \frac{55-62}{2} = -\frac{7}{2}^{\circ}\text{C}/\text{cm}$$

2-3-3

22.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0} = \frac{49-5}{80} = \frac{11}{20} ft/sec^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80)-v(0)}{80-0} = \frac{49-5}{80} = \frac{11}{20} ft/sec^2$$

2-3-3

23.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t=0$) and 8 P.M. ($t=8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

Use the data in the table to approximate the rate, in hundreds of entries per hour, at which entries were being deposited at time $t=6$. Show the computations that lead to your answer.

Part A

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$

2-3-3

24.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

Use the data in the table to approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer.

Part A

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$



0	1
---	---

The student response earns none of the following points:

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$

2-3-3

25.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

 Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

Part A

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$w'(12) \approx \frac{w(15) - w(9)}{15 - 9} = \frac{67.9}{6}$$

=1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t=12$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimate

1 point is earned for the interpretation with units

$$w'(12) \approx \frac{w(15) - w(9)}{15 - 9} = \frac{67.9}{6}$$

=1.017 (or 1.016)

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t=12$.

2-3-3

26.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a differentiable function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

Use the data in the table to approximate $C'(3.5)$. Show the computations that lead to your answer, and indicate units of measure.

Part A

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

2-3-3

27.

t (seconds)	0	8	20	25	32	40
$v(t)$ (meters per second)	3	5	-10	-8	-4	7

The velocity of a particle moving along the x -axis is modeled by a differentiable function v , where the position x is measured in meters, and time t is measured in seconds. Selected values of $v(t)$ are given in the table above. The particle is at position $x=7$ meters when $t=0$ seconds.

Estimate the acceleration of the particle at $t=36$ seconds. Show the computations that lead to your answer. Indicate units of measure.

Part A

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$



0	1
---	---

The student response earns one of the following points:

1 point is earned for the answer

$$a(36) = v'(36) \approx \frac{v(40) - v(32)}{40 - 32} = \frac{11}{8} \text{ meters/sec}^2$$

2-3-3

28.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

Use the data in the table to estimate the value of $v'(16)$.

Part A

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters}/\text{min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters}/\text{min}^2$$

2-3-3

29. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g , where $g(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Selected values of $g(t)$ are given in the table above.



(a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

Part B

2-3-3

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) \, dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) \, dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) \, dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) \, dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) \, dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) \, dt = 61.749$ (or 61.748) cubic feet.



0	1	2
---	---	---

The student response earns all of the following points:

2-3-3

1 point is earned for: integral**1 point is earned for:** answer

$$\int_0^8 w(t) dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) dt = 61.749$ (or 61.748) cubic feet.

Part D**1 point is earned for:** considers $g(6) - w(6)$ **1 point is earned for:** answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$ **1 point is earned for:** answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

2-3-3

30. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

t (minutes)	0	1	5	6	8
$g(t)$ (cubic feet per minute)	12.8	15.1	20.5	18.3	22.7

Grain is being added to a silo. At time $t = 0$, the silo is empty. The rate at which grain is being added is modeled by the differentiable function g , where $g(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Selected values of $g(t)$ are given in the table above.

 (a) Using the data in the table, approximate $g'(3)$. Using correct units, interpret the meaning of $g'(3)$ in the context of the problem.

(b) Write an integral expression that represents the total amount of grain added to the silo from time $t = 0$ to time $t = 8$. Use a right Riemann sum with the four subintervals indicated by the data in the table to approximate the integral.

(c) The grain in the silo is spoiling at a rate modeled by $w(t) = 32 \cdot \sqrt{\sin\left(\frac{\pi t}{74}\right)}$, where $w(t)$ is measured in cubic feet per minute for $0 \leq t \leq 8$ minutes. Using the result from part (b), approximate the amount of unspoiled grain remaining in the silo at time $t = 8$.

(d) Based on the model in part (c), is the amount of unspoiled grain in the silo increasing or decreasing at time $t = 6$? Show the work that leads to your answer.

Part A

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: approximation

1 point is earned for: interpretation with units

$$g'(3) \approx \frac{g(5) - g(1)}{5 - 1} = \frac{20.5 - 15.1}{4} = 1.35$$

At time $t = 3$ minutes, the rate at which grain is being added to the silo is increasing at a rate of 1.35 cubic feet per minute per minute.

2-3-3

Part B

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: integral expression

1 point is earned for: right Riemann sum

1 point is earned for: approximation

The total amount of grain added to the silo from time $t = 0$ to time $t = 8$ is $\int_0^8 g(t) dt$ cubic feet.

$$\begin{aligned}\int_0^8 g(t) dt &\approx g(1) \cdot (1 - 0) + g(5) \cdot (5 - 1) + g(6) \cdot (6 - 5) + g(8) \cdot (8 - 6) \\ &= 15.1 \cdot 1 + 20.5 \cdot 4 + 18.3 \cdot 1 + 22.7 \cdot 2 = 160.8\end{aligned}$$

Part C

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) dt = 61.749$ (or 61.748) cubic feet.



0	1	2
---	---	---

2-3-3

The student response earns all of the following points:

1 point is earned for: integral

1 point is earned for: answer

$$\int_0^8 w(t) \, dt = 99.051497$$

The approximate amount of unspoiled grain remaining in the silo at time $t = 8$ is $160.8 - \int_0^8 w(t) \, dt = 61.749$ (or 61.748) cubic feet.

Part D

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: considers $g(6) - w(6)$

1 point is earned for: answer

$$g(6) - w(6) = 18.3 - 16.063173 = 2.236827 > 0$$

Because $g(6) - w(6) > 0$, the amount of unspoiled grain is increasing at time $t = 6$.

2-3-3

31.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a differentiable function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table above.

(a) Use the data in the table to approximate $C'(3.5)$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t , $2 \leq t \leq 4$, at which $C'(t) = 2$? Justify your answer.

(c) Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\frac{1}{6} \int_0^6 C(t) dt$. Using correct units, explain the meaning of $\frac{1}{6} \int_0^6 C(t) dt$ in the context of the problem.

(d) The amount of coffee in the cup, in ounces, is modeled by $B(t) = 16 - 16e^{-0.4t}$. Using this model, find the rate at which the amount of coffee in the cup is changing when $t = 5$.

Part A

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the approximation

1 point is earned for units

$$C'(3.5) \approx \frac{C(4) - C(3)}{4 - 3} = \frac{12.8 - 11.2}{1} = 1.6 \text{ ounces/min}$$

Part B

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

2-3-3

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{C(4) - C(2)}{4 - 2}$

1 point is earned for the conclusion, using MVT

C is differentiable $\Rightarrow C$ is continuous (on the closed interval)

$$\frac{C(4) - C(2)}{4 - 2} = \frac{12.8 - 8.8}{2} = 2$$

Therefore, by the Mean Value Theorem, there is at least one time t , $2 < t < 4$, for which $C'(t) = 2$.

Part C

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

$$\begin{aligned} \frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces} \end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.

2-3-3



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the midpoint sum

1 point is earned for the approximation

1 point is earned for the interpretation

$$\begin{aligned}\frac{1}{6} \int_0^6 C(t) dt &\approx \frac{1}{6} [2 \cdot C(1) + 2 \cdot C(3) + 2 \cdot C(5)] \\ &= \frac{1}{6} (2 \cdot 5.3 + 2 \cdot 11.2 + 2 \cdot 13.8) \\ &= \frac{1}{6} (60.6) = 10.1 \text{ ounces}\end{aligned}$$

$\frac{1}{6} \int_0^6 C(t) dt$ is the average amount of coffee in the cup, in ounces, over the time interval $0 \leq t \leq 6$ minutes.

Part D

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$\begin{aligned}B'(t) &= -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t} \\ B'(5) &= 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}\end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $B'(t)$

1 point is earned for $B'(5)$

$$\begin{aligned}B'(t) &= -16(-0.4)e^{-0.4t} = 6.4e^{-0.4t} \\ B'(5) &= 6.4e^{-0.4(5)} = \frac{6.4}{e^2} \text{ ounces/min}\end{aligned}$$

2-3-3

32.



t (hours)	0	1	3	4	7	8	9
$L(t)$ (people)	120	156	176	126	150	80	0

Concert tickets went on sale at noon ($t = 0$) and were sold out within 9 hours. The number of people waiting in line to purchase tickets at time t is modeled by a twice-differentiable function L for $0 \leq t \leq 9$. Values of $L(t)$ at various times t are shown in the table above.

- (a) Use the data in the table to estimate the rate at which the number of people waiting in line was changing at 5:30 P.M. ($t = 5.5$). Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a trapezoidal sum with three subintervals to estimate the average number of people waiting in line during the first 4 hours that tickets were on sale.
- (c) For $0 \leq t \leq 9$, what is the fewest number of times at which $L'(t)$ must equal 0? Give a reason for your answer.
- (d) The rate at which tickets were sold for $0 \leq t \leq 9$ is modeled by $r(t) = 550te^{-t/2}$ tickets per hour. Based on the model, how many tickets were sold by 3 P.M. ($t = 3$), to the nearest whole number?

Part A

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for estimation

1 point is earned for units

$$L'(5.5) = \frac{L(7) - L(4)}{7 - 4} = \frac{150 - 126}{3} = 8 \text{ people per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

2-3-3

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for answer

The average number of people waiting in line during the first 4 hours is approximately

$$\frac{1}{4} \left(\frac{L(0) + L(1)}{2}(1 - 0) + \frac{L(1) + L(3)}{2}(3 - 1) + \frac{L(3) + L(4)}{2}(4 - 3) \right)$$

= 155.25 people

Part C

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

2-3-3



0	1	2	3
---	---	---	---

The student earns all of the following points:

1 point is earned for considering change in sign of L'

1 point is earned for the analysis

1 point is earned for the correct conclusion

L is differentiable $[0, 9]$ so the Mean Value Theorem implies $L'(t) > 0$ for some t in $(1, 3)$ and some t in $(4, 7)$. Similarly, $L'(t) < 0$ for some t in $(3, 4)$ and some t in $(7, 8)$. Then, since L' is continuous on $[0, 9]$, the Intermediate Value Theorem implies that $L'(t) = 0$ for at least three values of t in $[0, 9]$.

OR

1 point is earned for considering relative extrema of L on $(0, 9)$

1 point is earned for the analysis

1 point is earned for the correct conclusion

The continuity of L on $[1, 4]$ implies that L attains a maximum value there. Since $L(3) > L(1)$ and $L(3) > L(4)$, this maximum occurs on $(1, 4)$. Similarly, L attains a minimum on $(3, 7)$ and a maximum on $(4, 8)$. L is differentiable, so $L'(t) = 0$ at each relative extreme point on $(0, 9)$. Therefore $L'(t) = 0$ for at least three values of t in $[0, 9]$.

[Note: There is a function L that satisfies the given conditions with $L'(t) = 0$ for exactly three values of t .]

Part D

1 point is earned for the correct integrand

1 point is earned for the correct limits and answer

$$\int_0^3 r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.



0	1	2
---	---	---

The student earns all of the following points:

1 point is earned for the correct integrand

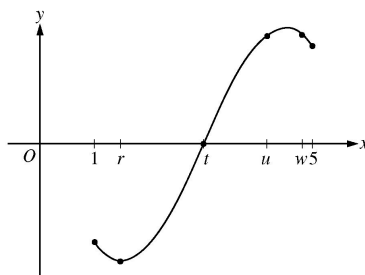
1 point is earned for the correct limits and answer

2-3-3

$$\int_0^3 r(t) dt = 972.784$$

There were approximately 973 tickets sold by 3 P.M.

33.

Graph of f

The figure above shows the graph of the differentiable function f for $1 \leq x \leq 5$. Which of the following could be the x -coordinate of a point at which the line tangent to the graph of f is parallel to the secant line through the points $(1, f(1))$ and $(5, f(5))$?

(A) r (B) t (C) u (D) w

(E) There is no such point.



2-3-3

34.

t (minutes)	0	2	5	9	10
$H(t)$ (degrees Celsius)	66	60	52	44	43

 As a pot of tea cools, the temperature of the tea is modeled by a differentiable function H for $0 \leq t \leq 10$, where time t is measured in minutes and temperature $H(t)$ is measured in degrees Celsius. Values of $H(t)$ at selected values of time t are shown in the table above.

(a) Use the data in the table to approximate the rate at which the temperature of the tea is changing at time $t = 3.5$. Show the computations that lead to your answer.

(b) Using correct units, explain the meaning of $\frac{1}{10} \int_0^{10} H(t) dt$ in the context of this problem. Use a trapezoidal sum with the four subintervals indicated by the table to estimate $\frac{1}{10} \int_0^{10} H(t) dt$.

(c) Evaluate $\int_0^{10} H'(t) dt$. Using correct units, explain the meaning of the expression in the context of this problem.

(d) At time $t = 0$, biscuits with temperature 100°C were removed from an oven. The temperature of the biscuits at time t is modeled by a differentiable function B for which it is known that $B'(t) = -13.84e^{-0.173t}$. Using the given models, at time $t = 10$, how much cooler are the biscuits than the tea?

Part A

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$H'(3.5) \approx \frac{H(5) - H(2)}{5 - 2}$$

$$= \frac{52 - 60}{3} = -2.666 \text{ or } -2.667 \text{ degrees Celsius per minute}$$

Part B

2-3-3

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\frac{1}{10} \int_0^{10} H(t) dt \approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ = 52.95$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for meaning of the expression

1 point is earned for trapezoidal sum

1 point is earned for estimate

$\frac{1}{10} \int_0^{10} H(t) dt$ is the average temperature of the tea, in degrees Celsius, over the 10 minutes.

$$\frac{1}{10} \int_0^{10} H(t) dt \approx \frac{1}{10} \left(2 \cdot \frac{66+60}{2} + 3 \cdot \frac{60+52}{2} + 4 \cdot \frac{52+44}{2} + 1 \cdot \frac{44+33}{2} \right) \\ = 52.95$$

Part C

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

2-3-3



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the value of integral

1 point is earned for meaning of the expression

$$\int_0^{10} H'(t) dt = H(10) - H(0) = 43 - 66 = -23$$

The temperature of the tea drops 23 degrees Celsius from time $t = 0$ to time $t = 10$ minutes.

Part D

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer $B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275$; $H(10) - H(0) = 8.817$

The biscuits are 8.817 degrees Celsius cooler than the tea.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for using $B(0) = 100$

1 point is earned for answer

$$B(10) = 100 + \int_0^{10} B'(t) dt = 34.18275; H(10) - H(0) = 8.817$$

The biscuits are 8.817 degrees Celsius cooler than the tea.

2-3-3

35.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$g(x)$	$g'(x)$
2	-2	4
4	5	1
6	2	-3

The table above gives values of the differentiable function g and its derivative g' at selected values of x . Let f be the function defined by $f(x) = x \sin(\sqrt{3x^2 + 5})$.

- Using values of g from the table, estimate $g'(5)$. Show the work that leads to your answer.
- Let R be the function defined by $R(x) = f(x) - g(x)$. Find $R'(2)$.
- Let S be the function defined by $S(x) = f(x)g(x)$. Find $S'(4)$.
- Let T be the function defined by $T(x) = \frac{f(x)}{g(x)}$. Find $T'(6)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct estimate of $g'(5)$

Solution:

2-3-3

$$g'(5) \approx \frac{g(6)-g(4)}{6-4} = \frac{2-5}{6-4} = -\frac{3}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a difference, either by writing $R'(x) = f'(x) - g'(x)$ or $R'(2) = f'(2) - g'(2)$ or $R'(2) = -2.448854 - 4$ OR for the correct answer accompanied by $f'(2) = -2.448854$ and $g'(2) = 4$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $-2.448854 - 4$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $R'(x)$
- ☐ answer

Solution:

$$R'(x) = f'(x) - g'(x)$$

$$f'(2) = -2.448854 \text{ from the calculator.}$$

$$g'(2) = 4 \text{ from the table.}$$

$$R'(2) = f'(2) - g'(2) = -6.449$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $S'(x) = f'(x)g(x) + f(x)g'(x)$ or $S'(4) = f'(4)g(4) + f(4)g'(4)$ or

$$S'(4) = (4.419230) \cdot (5) + (3.359221) \cdot (1).$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$$(4.419230) \cdot (5) + (3.359221) \cdot (1) \text{ earns the second point.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-3-3



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $S'(x)$
- ☐ answer

Solution:

$$S'(x) = f'(x)g(x) + f(x)g'(x)$$

$f(4) = 3.359221$ and $f'(4) = 4.419230$ from the calculator.

$g(4) = 5$ and $g'(4) = 1$ from the table.

$$S'(4) = f'(4)g(4) + f(4)g'(4) = 25.455$$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \text{ or } T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} \text{ or}$$

$$T'(6) = \frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as

$$\frac{(-4.564565) \cdot (2) - (-5.603824) \cdot (-3)}{(2)^2} \text{ earns the second point.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $T'(x)$
- ☐ answer

Solution:

2-3-3

$$T'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$$

$$f(6) = -5.603824 \text{ and}$$

$$f'(6) = -4.564565 \text{ from the calculator.}$$

$$g(6) = 2 \text{ and } g'(6) = -3 \text{ from the table.}$$

$$T'(6) = \frac{f'(6)g(6) - f(6)g'(6)}{(g(6))^2} = -6.485$$

2-3-3

36.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

x	$f(x)$	$f'(x)$
3	-1	8
5	4	3
7	6	-2

The table above gives values of the differentiable function f and its derivative f' at selected values of x . Let h be the function defined by $h(x) = x^2 e^{\cos(3x)}$.

- Using values of f from the table, estimate $f'(4)$. Show the work that leads to your answer.
- Let M be the function defined by $M(x) = f(x) + h(x)$. Find $M'(3)$.
- Let N be the function defined by $N(x) = f(x)h(x)$. Find $N'(5)$.
- Let P be the function defined by $P(x) = \frac{h(x)}{f(x)}$. Find $P'(7)$.

Part A

The point is earned for a correct expression that substitutes correct values from the table. The expression does not need to be simplified. If simplified, there are no errors. A response that presents an incorrect expression, makes an error in substituting values from the table, or makes an error in simplification does not earn the point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct estimate of $f'(4)$

Solution:

2-3-3

$$f'(4) \approx \frac{f(5)-f(3)}{5-3} = \frac{4-(-1)}{5-3} = \frac{5}{2}$$

Part B

The first point is earned for correctly presenting the rule for derivative of a sum, either by writing $M'(x) = f'(x) + h'(x)$ or $M'(3) = f'(3) + h'(3)$ or $M'(3) = 8 + (-2.061490)$ OR for the correct answer accompanied by $f'(3) = 8$ and $h'(3) = -2.061490$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $8 + (-2.061490)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $M'(x)$
- ☐ answer

Solution:

$$M'(x) = f'(x) + h'(x)$$

$$f'(3) = 8 \text{ from the table.}$$

$$h'(3) = -2.061490 \text{ from the calculator.}$$

$$M'(3) = f'(3) + h'(3) = 5.939 \text{ (or } 5.938)$$

Part C

The first point is earned for correctly presenting the product rule for derivatives, either by writing $N'(x) = f'(x)h(x) + f(x)h'(x)$ or $N'(5) = f'(5)h(5) + f(5)h'(5)$ or $N'(5) = 3(11.695310) + 4(-18.137830)$.

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $3(11.695310) + 4(-18.137830)$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

2-3-3

The student response accurately includes both of the criteria below.

- ☐ $N'(x)$
- ☐ answer

Solution:

$$N'(x) = f'(x)h(x) + f(x)h'(x)$$

$f(5) = 4$ and $f'(5) = 3$ from the table.

$h(5) = 11.695310$ and $h'(5) = -18.137830$ from the calculator.

$$N'(5) = f'(5)h(5) + f(5)h'(5) = -37.465$$

Part D

The first point is earned for correctly presenting the quotient rule for derivatives, either by writing

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2} \text{ or } P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} \text{ or } P'(7) = \frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}.$$

To be eligible for the second point, the response must have earned the first point or made a maximum of one sign error with either term in the numerator and correctly evaluated the expression. An answer without supporting work does not earn any points. An unsimplified final answer such as $\frac{(-63.023773) \cdot 6 - 28.334809 \cdot (-2)}{6^2}$ earns the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $P'(x)$
- ☐ answer

Solution:

$$P'(x) = \frac{h'(x)f(x) - h(x)f'(x)}{(f(x))^2}$$

$f(7) = 6$ and $f'(7) = -2$ from the table.

$h(7) = 28.334809$ and $h'(7) = -63.023773$ from the calculator.

$$P'(7) = \frac{h'(7)f(7) - h(7)f'(7)}{(f(7))^2} = -8.930 \text{ (or } -8.929)$$

2-3-3

37.

t (hours)	0	2	5	7	8
$E(t)$ (hundreds of entries)	0	4	13	21	23

A zoo sponsored a one-day contest to name a new baby elephant. Zoo visitors deposited entries in a special box between noon ($t = 0$) and 8 P.M. ($t = 8$). The number of entries in the box t hours after noon is modeled by a differentiable function E for $0 \leq t \leq 8$. Values of $E(t)$, in hundreds of entries, at various times t are shown in the table above.

(a) Use the data in the table to approximate the rate, in hundreds of entries per hour, at which entries were being deposited at time $t = 6$. Show the computations that lead to your answer.

(b) Use a trapezoidal sum with the four subintervals given by the table to approximate the value of $18 \int_0^8 E(t) dt$.

. Using correct units, explain the meaning of $18 \int_0^8 E(t) dt$ in terms of the number of entries.

(c) At 8 P.M., volunteers began to process the entries. They processed the entries at a rate modeled by the function P , where $P(t) = t^3 - 30t^2 + 298t - 976$ hundreds of entries per hour for $8 \leq t \leq 12$. According to the model, how many entries had not yet been processed by midnight ($t = 12$)?

(d) According to the model from part (c), at what time were the entries being processed most quickly? Justify your answer.

Part A

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for answer

$$E'(6) \approx \frac{E(7) - E(5)}{7 - 5} = 4 \text{ hundred entries per hour}$$

Part B

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

2-3-3

$$\frac{1}{8} \int_0^8 E(t) dt \approx$$

$$\frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right)$$

$$= 10.687 \text{ or } 10.688$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for trapezoidal sum

1 point is earned for approximation

1 point is earned for meaning

$$\frac{1}{8} \int_0^8 E(t) dt \approx$$

$$\frac{1}{8} \left(2 \cdot \frac{E(0)+E(2)}{2} + 3 \cdot \frac{E(2)+E(5)}{2} + 2 \cdot \frac{E(5)+E(7)}{2} + 1 \cdot \frac{E(7)+E(8)}{2} \right)$$

$$= 10.687 \text{ or } 10.688$$

$$\frac{1}{8} \int_0^8 E(t) dt \text{ is the average number of hundreds of entries in the box between noon and 8 P.M.}$$

Part C

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$



0	1	2
---	---	---

2-3-3

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$23 - \int_8^{12} P(t) dt = 23 - 16 = 7 \text{ hundred entries}$$

Part D

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $P'(t) = 0$

1 point is earned for identifying candidates

1 point is earned for answering with justification

$P'(t) = 0$ when $t = 9.183503$ and $t = 10.816497$.

t	$P(t)$
8	0
9.183503	5.088662
10.816497	2.911338
12	8

Entries are being processed most quickly at time $t = 12$.

2-3-3

38.  Grass clippings are placed in a bin, where they decompose. For $0 \leq t \leq 30$, the amount of grass clippings remaining in the bin is modeled by $A(t) = 6.687(0.931)^t$, where $A(t)$ is measured in pounds and t is measured in days.
- (a) Find the average rate of change of $A(t)$ over the interval $0 \leq t \leq 30$. Indicate units of measure.
- (b) Find the value of $A'(15)$. Using correct units, interpret the meaning of the value in the context of the problem.
- (c) Find the time t for which the amount of grass clippings in the bin is equal to the average amount of grass clippings in the bin over the interval $0 \leq t \leq 30$.
- (d) For $t > 30$, $L(t)$, the linear approximation to A at $t = 30$, is a better model for the amount of grass clippings remaining in the bin. Use $L(t)$ to predict the time at which there will be 0.5 pound of grass clippings remaining in the bin. Show the work that leads to your answer.

Part A

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer with units

$$\frac{A(30) - A(0)}{30 - 0} = -0.197 \text{ (or } -0.196) \text{ lbs/day}$$

Part B

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $A'(15)$

1 point is earned for the interpretation

$$A'(15) = -0.164 \text{ (or } -0.163)$$

2-3-3

The amount of grass clippings in the bin is decreasing at a rate of 0.164 (or 0.163) lbs/day at time $t = 15$ days.

Part C

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $\frac{1}{30} \int_0^{30} A(t) dt$

1 point is earned for the answer

$$A(t) = \frac{1}{30} \int_0^{30} A(t) dt \Rightarrow t = 12.415 \text{ (or 12.414)}$$

Part D

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$

2-3-3



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for expression for $L(t)$

1 point is earned for $L(t) = 0.5$

1 point is earned for the answer

$$L(t) = A(30) + A'(30) \cdot (t - 30)$$

$$A'(30) = -0.055976$$

$$A(30) = 0.782928$$

$$L(t) = 0.5 \Rightarrow t = 35.054$$

2-3-3

39.

Distance x (cm)	0	1	5	6	8
Temperature $T(x)$ ($^{\circ}\text{C}$)	100	93	70	62	55

A metal wire of length 8 centimeters (cm) is heated at one end. The table above gives selected values of the temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), of the wire x cm from the heated end. The function T is decreasing and twice differentiable.

- (a) Estimate $T'(7)$. Show the work that leads to your answer. Indicate units of measure.
- (b) Write an integral expression in terms of $T(x)$ for the average temperature of the wire. Estimate the average temperature of the wire using a trapezoidal sum with the four subintervals indicated by the data in the table. Indicate units of measure.
- (c) Find $\int_0^8 T'(x)dx$, and indicate units of measure. Explain the meaning of $\int_0^8 T'(x)dx$ in terms of the temperature of the wire.
- (d) Are the data in the table consistent with the assertion that $T''(x) > 0$ for every x in the interval $0 < x < 8$? Explain your answer.

Part A

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\frac{T(8) - T(6)}{8 - 6} = \frac{55 - 62}{2} = -\frac{7}{2}^{\circ}\text{C/cm}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C/cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part B

2-3-3

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

$$\text{Average temperature} \approx \frac{1}{8} A = 75.6875^\circ \text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^\circ\text{C}/\text{cm}$ in (a), and $^\circ\text{C}$ in (b) and (c)



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for: $\frac{1}{8} \int_0^8 T(x) dx$

1 point is earned for the trapezoidal sum

1 point is earned for the answer

$$\frac{1}{8} \int_0^8 T(x) dx$$

Trapezoidal approximation for $\int_0^8 T(x) dx$:

$$A = \frac{100 + 93}{2} \cdot 1 + \frac{93 + 70}{2} \cdot 4 + \frac{70 + 62}{2} \cdot 1 + \frac{62 + 55}{2} \cdot 2$$

2-3-3

$$\text{Average temperature} \approx \frac{1}{8}A = 75.6875^{\circ}\text{C}$$

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part C

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the value

1 point is earned for the meaning

$$\int_0^8 T'(x)dx = T(8) - T(0) = 55 - 100 = -45^{\circ}\text{C}$$

The temperature drops 45°C from the heated end of the wire to the other end of the wire.

1 point is earned for: units in (a), (b), and (c)

Units of $^{\circ}\text{C}/\text{cm}$ in (a), and $^{\circ}\text{C}$ in (b) and (c)

Part D

1 point is earned for the two slopes of secant lines

2-3-3

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the two slopes of secant lines

1 point is earned for the answer with explanation

Average rate of change of temperature on $[1, 5]$ is $\frac{70 - 93}{5 - 1} = -5.75$.

Average rate of change of temperature on $[5, 6]$ is $\frac{62 - 70}{6 - 5} = -8$.

No. By the MVT, $T'(c_1) = -5.75$ for some c_1 in the interval $(1, 5)$ and $T'(c_2) = -8$ for some c_2 in the interval $(5, 6)$. It

follows that T' must decrease somewhere in the interval (c_1, c_2) . Therefore T'' is not positive for every x in $[0, 8]$.

2-3-3

40.

t (minutes)	0	12	20	24	40
$v(t)$ (meters per minute)	0	200	240	-220	150

Johanna jogs along a straight path. For $0 \leq t \leq 40$, Johanna's velocity is given by a differentiable function v . Selected values of $v(t)$, where t is measured in minutes and $v(t)$ is measured in meters per minute, are given in the table above.

(a) Use the data in the table to estimate the value of $v'(16)$.

(b) Using correct units, explain the meaning of the definite integral $\int_0^{40} |v(t)| dt$ in the context of the problem.

Approximate the value of $\int_0^{40} |v(t)| dt$ using a right Riemann sum with the four subintervals indicated in the table.

(c) Bob is riding his bicycle along the same path. For $0 \leq t \leq 10$, Bob's velocity is modeled by $B(t) = t^3 - 6t^2 + 300$, where t is measured in minutes and $B(t)$ is measured in meters per minute. Find Bob's acceleration at time $t = 5$.

(d) Based on the model B from part (c), find Bob's average velocity during the interval $0 \leq t \leq 10$.

Part A

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the approximation

$$v'(16) \approx \frac{240-200}{20-12} = 5 \text{ meters/min}^2$$

Part B

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

2-3-3

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned for the right Riemann sum

1 point is earned for the approximation

$\int_0^{40} |v(t)| dt$ is the total distance Johanna jogs, in meters, over the time interval $0 \leq t \leq 40$ minutes.

$$\int_0^{40} |v(t)| dt \approx 12 \cdot |v(12)| + 8 \cdot |v(20)| + 4 \cdot |v(24)| + 16 \cdot |v(40)|$$

$$= 12 \cdot 200 + 8 \cdot 240 + 4 \cdot 220 + 16 \cdot 150$$

$$= 2400 + 1920 + 880 + 2400$$

$$= 7600 \text{ meters}$$

Part C

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for using $B'(t)$

1 point is earned for the answer

Bob's acceleration is $B'(t) = 3t^2 - 12t$.

$$B'(5) = 3(25) - 12(5) = 15 \text{ meters/min}^2$$

2-3-3

Part D

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}\end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned}\text{Avg vel} &= \frac{1}{10} \int_0^{10} (t^3 - 6t^2 + 300) dt \\ &= \frac{1}{10} \left[\frac{t^4}{4} - 2t^3 + 300t \right]_0^{10} \\ &= \frac{1}{10} \left[\frac{10000}{4} - 2000 + 3000 \right] = 350 \text{ meters/min}\end{aligned}$$

2-3-3

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	5	8	10
$G(t)$ (thousands of gallons)	22	18.5	19.5	22	24

A gasoline storage tank is filled by a pipeline from a refinery. At the same time, gasoline flows from the tank into trucks that will make deliveries to gasoline stations. The amount of gasoline in the storage tank at time t is given by the twice-differentiable function G , where t is measured in hours and $0 \leq t \leq 10$. Values of $G(t)$, in thousands of gallons, at selected times t are given in the table above. It is known that $G''(t) > 0$ for $5 \leq t < 10$.

- Use the data in the table to estimate the rate of change of the amount of gasoline in the storage tank at time $t = 3$ hours. Show the computations that lead to your answer. Indicate units of measure.
- For $0 < t < 10$, is there a time t at which $G(t)$ is increasing at a rate of 0.2 thousand gallons per hour? Justify your answer.
- It is known that $G'(5) = 0.5$. Use the locally linear approximation for G at time $t = 5$ to approximate the amount of gasoline in the storage tank at time $t = 7$. Is this approximation an overestimate or an underestimate for the actual amount of gasoline in the storage tank at time $t = 7$? Give a reason for your answer.
- The rate at which gasoline flows out of the storage tank into trucks at time t can be modeled by the function R defined by $R(t) = \frac{100t}{t^2+4}$, where t is measured in hours, and $R(t)$ is measured in thousands of gallons. Based on the model, at what time t , for $0 \leq t \leq 10$, is the rate at which gasoline flows out of the storage tank an absolute maximum? Justify your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation with units.

Solution:

$$G'(3) \approx \frac{G(5) - G(1)}{5 - 1} = \frac{19.5 - 18.5}{5 - 1} = \frac{1}{4} = 0.25 \text{ thousand gallons per hour (which is equivalent to 250 gallons per hour)}$$

Part B

2-3-3

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{G(10)-G(0)}{10-0}$
- ☐ justification using Mean Value Theorem

Solution:

G is twice differentiable. $\Rightarrow G$ is differentiable. $\Rightarrow G$ is continuous.

$$\frac{G(10)-G(0)}{10-0} = \frac{24-22}{10-0} = \frac{2}{10} = 0.2 \text{ thousand gallons per hour (which is equivalent to 200 gallons per hour)}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 < t < 10$, at which $G'(t) = 0.2$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation of $G(7)$
- ☐ underestimate with reason

Solution:

$$G(7) \approx G(5) + G'(5)(7-5) = 19.5 + (0.5)(2) = 20.5 \text{ thousand gallons}$$

$G''(t) > 0$ for $5 \leq t \leq 7$, and so the graph of G is concave up for $5 \leq t \leq 7$. Therefore, the approximation for $G(7)$ using the locally linear approximation for G at time $t = 5$ is an underestimate for the actual amount.

Part D

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

2-3-3

The response is eligible for additional points based on consistent answers using an incorrect $R'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ $R'(t)$
- ☐ sets $R'(t) = 0$
- ☐ identifies $t = 2$ as a candidate
- ☐ answer with justification

Solution:

$$R'(t) = \frac{100(t^2+4) - 100t(2t)}{(t^2+4)^2} = \frac{400 - 100t^2}{(t^2+4)^2} = \frac{100(4-t^2)}{(t^2+4)^2} = 0$$

$$\Rightarrow 4 - t^2 = 0 \Rightarrow t = 2$$

Because $R'(t)$ changes from positive to negative at $t = 2$, R has a relative maximum at $t = 2$. R is increasing on $0 \leq t \leq 2$ and R is decreasing on $2 \leq t \leq 10$. Since $t = 2$ is the only critical point of R on the interval $0 \leq t \leq 10$ that is the location of a relative maximum, it is also the location of the absolute maximum of R on the interval $0 \leq t \leq 10$.

2-3-3

42.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	3	8	12
$H(t)$ (degrees Celsius)	90	80	67	58

As a bowl of soup cools, the temperature of the soup is given by the twice-differentiable function H for $0 \leq t \leq 12$, where $H(t)$ is measured in degrees Celsius ($^{\circ}\text{C}$) and time t is measured in minutes. Values of $H(t)$ at selected values of time t are shown in the table above.

- Use the data in the table to approximate $H'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $H'(5)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 12$, at which the temperature of the soup is 60°C ? Justify your answer.
- The temperature of the soup is also modeled by the twice-differentiable function C for $0 \leq t \leq 12$, where $C(t)$ is measured in degrees Celsius and time t is measured in minutes. It is known that $C(3) = 80$ and the derivative of C is given by $C'(t) = -3.6e^{-0.05t}$. Write an equation for the line tangent to the graph of C at $t = 3$ and use it to approximate $C(5)$.
- Based on the model given in part (c), at what rate, in degrees Celsius per minute per minute, is the rate of change of the temperature of the soup changing at time $t = 3$ minutes?

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-3-3



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$H'(5) \approx \frac{H(8) - H(3)}{8 - 3} = \frac{67 - 80}{5} = -2.6$$

At time $t = 5$ minutes, the temperature of the soup is changing at a rate of -2.6 degrees Celsius per minute.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ H is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

H is twice differentiable. $\Rightarrow H$ is continuous.

$$H(12) = 58 < 60 < 90 = H(0)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 12$, such that $H(t) = 60$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $C'(3)$.

The third point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-3-3



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ slope
- ☐ equation for tangent line
- ☐ approximation

Solution:

$$C'(3) = -3.098549$$

An equation for the tangent line is $y = 80 - 3.098549(t - 3)$.

$$C(5) \approx 80 - C'(3)(5 - 3) = 73.802903$$

$$C(5) = 73.803 \text{ (or } 73.802)$$

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $C''(3)$
- ☐ interpretation with units

Solution:

$$C''(3) = 0.154927$$

At time $t = 3$ minutes, the rate of change of the temperature is changing by 0.155 (or 0.154) degree Celsius per minute per minute.

2-3-3

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	5	15	30	35
$A(t)$ (gallons)	10	18	25	16	8

The number of gallons of olive oil in a tank at time t is given by the twice-differentiable function A , where t is measured in hours and $0 \leq t \leq 35$. Values of $A(t)$ at selected times t are given in the table above.

- (a) Use the data in the table to estimate the rate at which the number of gallons of olive oil in the tank is changing at time $t = 10$ hours. Show the computations that lead to your answer. Indicate units of measure.
- (b) For $0 \leq t \leq 30$, is there a time t at which $A'(t) = \frac{1}{5}$? Justify your answer.
- (c) The number of gallons of olive oil in the tank at time t is also modeled by the function G defined by $G(t) = 5t - \frac{2}{3}(t+9)^{\frac{3}{2}} + 28$, where t is measured in hours and $0 \leq t \leq 35$. Based on the model, at what time t , for $0 \leq t \leq 35$, is the number of gallons of olive oil in the tank an absolute maximum? Justify your answer.
- (d) For the function G defined in part (c), the locally linear approximation near $t = 7$ is used to approximate $G(8)$. Is this approximation an overestimate or an underestimate for the value of $G(8)$? Give a reason for your answer.

Part A

Substitution of function values is required. The answer does not need to be simplified.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes the criteria below.

- ☐ approximation with units

Solution:

$$A'(10) \approx \frac{A(15) - A(5)}{15 - 5} = \frac{25 - 18}{15 - 5} = \frac{7}{5} \text{ gallons per hour}$$

Part B

2-3-3

The first point requires identification of the difference quotient. The substitution of function values and conclusion impacts the second point.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $\frac{A(30)-A(0)}{30-0}$
- ☐ justification using Mean Value Theorem

Solution:

A is twice differentiable. $\Rightarrow A$ is differentiable. $\Rightarrow A$ is continuous.

$$\frac{A(30)-A(0)}{30-0} = \frac{16-10}{30-0} = \frac{6}{30} = \frac{1}{5}$$

Therefore, by the Mean Value Theorem, there is a time t , for $0 \leq t \leq 30$, at which $A'(t) = \frac{1}{5}$.

Part C

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The response is eligible for additional points based on consistent answers using an incorrect $G'(t)$ with a maximum of one computational error.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ $G'(t)$
- ☐ sets $G'(t) = 0$
- ☐ identifies $t = 16$ as a candidate
- ☐ answer with justification

Solution:

2-3-3

$$G'(t) = 5 - (t + 9)^{\frac{1}{2}} = 0 \Rightarrow (t + 9)^{\frac{1}{2}} = 5$$

$$\Rightarrow t + 9 = 25 \Rightarrow t = 16$$

Because $G'(t)$ changes from positive to negative at $t = 16$, G has a relative maximum at $t = 16$.

Since $t = 16$ is the only critical point of G on the interval $0 \leq t \leq 35$ that is the location of a relative maximum, it is also the location of the absolute maximum of G on the interval $0 \leq t \leq 35$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $G''(t)$
- ☐ overestimate with reason

Solution:

$$G''(t) = -\frac{1}{2}(t + 9)^{-\frac{1}{2}}$$

$G''(t) < 0$ for $7 \leq t \leq 8$, and so the graph of G is concave down for $7 \leq t \leq 8$. Therefore, the approximation of $G(8)$ using the locally linear approximation of G near $t = 7$ is an overestimate.

2-3-3

44. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	3	8	12	14
$S'(t)$ (centimeters per hour)	2.3	2.1	1.9	1.7	1.6

The depth of snow in a field is given by the twice-differentiable function S for $0 \leq t \leq 14$, where $S(t)$ is measured in centimeters and time t is measured in hours. Values of $S'(t)$, the derivative of S , at selected values of time t are shown in the table above. It is known that the graph of S is concave down for $0 \leq t \leq 14$.

- Use the data in the table to approximate $S''(10)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $S''(10)$ in the context of the problem.
- Is there a time t , for $0 \leq t \leq 14$, at which the depth of snow is changing at a rate of 2 centimeters per hour? Justify your answer.
- At time $t = 8$, the depth of snow is 45 centimeters. Use the line tangent to the graph of S at $t = 8$ to approximate the depth of snow at time $t = 10$. Is the approximation an underestimate or an overestimate of the actual depth of snow at time $t = 10$? Justify your answer.
- The depth of snow, in centimeters, is also modeled by the twice-differentiable function D for $0 \leq t \leq 14$, where $D(t) = 120 - 92e^{-\frac{t}{40}}$ and time t is measured in hours. Use the model to find $D'(10)$. Using correct units, explain the meaning of $D'(10)$ in the context of the problem.

Part A

The first point requires a correct substitution of values. A simplified answer is not required.

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-3-3



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ approximation
- ☐ interpretation with units

Solution:

$$S''(10) \approx \frac{S'(12) - S'(8)}{12 - 8} = \frac{1.7 - 1.9}{4} = -0.05$$

At time $t = 10$ hours, the rate of change of the depth of snow is changing at a rate of -0.05 centimeter per hour per hour.

Part B

The second point may be earned if response contains conditions of IVT without naming the theorem.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ S' is continuous
- ☐ justification using Intermediate Value Theorem

Solution:

S is twice differentiable. $\Rightarrow S'$ is differentiable and thus continuous.

$$S'(8) = 1.9 < 2 < 2.1 = S'(3)$$

By the Intermediate Value Theorem, there is a value of t , for $0 \leq t \leq 14$, such that $S'(t) = 2$.

Part C

The second point may be earned if using an incorrect value for slope, provided the value is labeled as $S'(8)$.

The second point does not require a simplified answer. Substitution of function values is required.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

2-3-3



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ equation for tangent line
- ☐ approximation
- ☐ overestimate with justification using concavity

Solution:

$$S'(8) = 1.9 \text{ from table}$$

An equation for the tangent line is $y = 45 + 1.9(t - 8)$.

$$S(10) \approx 45 + S'(8)(10 - 8) = 48.8 \text{ centimeters}$$

This approximation is an overestimate because the graph of S is concave down for $8 \leq t \leq 10$.

Part D

The second point requires a time reference, rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $D'(10)$
- ☐ interpretation with units

Solution:

$$D'(10) = 1.791$$

At time $t = 10$ hours, the depth of snow is changing by 1.791 centimeters per hour.

2-3-3

45. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	2	5	9	11	12
$L(t)$ (cars per hour)	15	40	24	68	18

The rate at which cars enter a parking lot is modeled by $E(t) = 30 + 5(t - 2)(t - 5)e^{-0.2t}$. The rate at which cars leave the parking lot is modeled by the differentiable function L . Selected values of $L(t)$ are given in the table above. Both $E(t)$ and $L(t)$ are measured in cars per hour, and time t is measured in hours after 5 A.M. ($t = 0$). Both functions are defined for $0 \leq t \leq 12$.

 (a) What is the rate of change of $E(t)$ at time $t = 7$? Indicate units of measure.

(b) How many cars enter the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole number.

(c) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_2^{12} L(t) dt$.

Using correct units, explain the meaning of $\int_2^{12} L(t) dt$ in the context of this problem.

(d) For $0 \leq t < 6$, 5 dollars are collected from each car entering the parking lot. For $6 \leq t \leq 12$, 8 dollars are collected from each car entering the parking lot. How many dollars are collected from the cars entering the parking lot from time $t = 0$ to time $t = 12$? Give your answer to the nearest whole dollar.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0

1

The student response accurately includes the criteria below.

2-3-3

☐ answer with units

Solution:

$$E'(7) = 6.164924$$

The rate of change of $E(t)$ at time $t = 7$ is 6.165 (or 6.164) cars per hour per hour.

Part B

The first point is earned with $E(t)$ or $30 + 5(t - 2)(t - 5)e^{-0.2t}$ written as the integrand. Limits are part of the second point.

It is possible to earn the second point with a single copy error of a constant in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ integrand

☐ answer

Solution:

$$\int_0^{12} E(t) dt = 520.070489$$

To the nearest whole number, 520 cars enter the parking lot from time $t = 0$ to time $t = 12$.

Part C

The first point is earned for showing the method of a trapezoidal sum, so the sum of three products must be given. This point can be earned with a maximum of one error in either a coefficient or a value extracted from the table.

The second point can be earned concurrently with the first point. Simplification is not required. Functions values from the table must be substituted.

The third point requires that the explanation includes (1) units (cars), (2) the time interval ($2 \leq t \leq 12$ or 7 A.M. to 5 P.M.), (3) a meaning that expresses the (approximate) number of cars leaving the parking lot.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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2-3-3

The student response accurately includes **all three** of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ explanation

Solution:

$$\begin{aligned}\int_2^{12} L(t) dt &\approx (5-2) \cdot \frac{L(2)+L(5)}{2} + (9-5) \cdot \frac{L(5)+L(9)}{2} \\ &\quad + (11-9) \cdot \frac{L(9)+L(11)}{2} + (12-11) \cdot \frac{L(11)+L(12)}{2} \\ &= 3 \cdot \frac{15+40}{2} + 4 \cdot \frac{40+24}{2} + 2 \cdot \frac{24+68}{2} + 1 \cdot \frac{68+18}{2} \\ &= 345.5\end{aligned}$$

$\int_2^{12} L(t) dt$ is the number of cars that leave the parking lot in the 10 hours between 7 A.M. ($t = 2$) and 5 P.M. ($t = 12$).

Part D

The first point is earned with two definite integrals, both having $E(t)$ as the integrand.

The third point may be earned with an answer correctly rounded or truncated to 1, 2, or 3 decimal places.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ integrand
- ☐ limits and constants
- ☐ answer

Solution:

$$5 \int_0^6 E(t) dt + 8 \int_6^{12} E(t) dt = 3530.1396$$

To the nearest dollar, 3530 dollars are collected from time $t = 0$ to time $t = 12$.

2-3-3

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (days)	0	3	7	10	12
$r'(t)$ (centimeters per day)	-6.1	-5.0	-4.4	-3.8	-3.5

An ice sculpture melts in such a way that it can be modeled as a cone that maintains a conical shape as it decreases in size. The radius of the base of the cone is given by a twice-differentiable function r , where $r(t)$ is measured in centimeters and t is measured in days. The table shown gives selected values of $r'(t)$, the rate of change of the radius, over the time interval $0 \leq t \leq 12$.

(a) Approximate $r''(8.5)$ using the average rate of change of r' over the interval $7 \leq t \leq 10$. Show the computations that lead to your answer, and indicate units of measure.

(b) Is there a time t , $0 \leq t \leq 3$, for which $r'(t) = -6$? Justify your answer.

(c) Use a right Riemann sum with the four subintervals indicated in the table to approximate the value of $\int_0^{12} r'(t) \, dt$.

(d) The height of the cone decreases at a rate of 2 centimeters per day. At time $t = 3$ days, the radius is 100 centimeters and the height is 50 centimeters. Find the rate of change of the volume of the cone with respect to time, in cubic centimeters per day, at time $t = 3$ days. (The volume V of a cone with radius r and height h is $V = \frac{1}{3}\pi r^2 h$.)

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the supporting work must include at least a difference and a quotient.

Simplification is not required to earn the first point. If the numerical value is simplified, it must be correct.

The second point can be earned with an incorrect approximation for $r''(8.5)$ but cannot be earned without some value for $r''(8.5)$ presented.

Units may be written in any equivalent form (such as cm/day^2).

2-3-3



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $r''(8.5)$ with supporting work
- ☐ Units

Solution:

$$r''(8.5) \approx \frac{r'(10) - r'(7)}{10 - 7} = \frac{-3.8 - (-4.4)}{10 - 7} \text{ centimeter per day per day}$$

$$= \frac{0.6}{3} = 0.2$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, the response must establish that -6 is between $r'(0)$ and $r'(3)$ (or -6.1 and -5). This statement may be represented symbolically (with or without including one or both endpoints in an inequality) or verbally. A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not state that -6 is between -6.1 and -5 . Thus this response does not earn the first point.

To earn the second point:

- The response must state that $r'(t)$ is continuous because $r'(t)$ is differentiable (or because $r(t)$ is twice differentiable).
- The response must have earned the first point. Exception: A response of “ $r'(t) = -6$ because $r'(0) = -6.1$ and $r'(3) = -5$ ” does not earn the first point because of imprecise communication but may nonetheless earn the second point if all other criteria for the second point are met.
- The response must conclude that there is a time t such that $r'(t) = -6$ (A statement of “yes” would be sufficient.)

To earn the second point, the response need not explicitly name the Intermediate Value Theorem, but if a theorem is named, it must be correct.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $r'(0) < -6 < r'(3)$
- ☐ Conclusion using Intermediate Value Theorem

Solution:

2-3-3

$r(t)$ is twice-differentiable. $\Rightarrow r'(t)$ is differentiable.

$\Rightarrow r'(t)$ is continuous.

$$r'(0) = -6.1 < -6 < -5.0 = r'(3)$$

Therefore, by the Intermediate Value Theorem, there is a time t , $0 < t < 3$, such that $r'(t) = -6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, at least seven of the eight factors in the Riemann sum must be correct. If there is any error in the Riemann sum, the response does not earn the second point.

A response of $3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5)$ earns both the first and second points, unless there is a subsequent error in simplification, in which case the response would earn only the first point.

A response that presents the correct answer, with accompanying work that shows the four products in the Riemann sum (without explicitly showing all of the factors and/or the sum process) does not earn the first point but earns the second point. For example, $-15 + 4(-4.4) + 3(-3.8) + -7$ does not earn the first point but earns the second point. Similarly, $-15, -17.6, -11.4, -7 \rightarrow -51$ does not earn the first point but earns the second point.

A response that presents the correct answer (-51) with no supporting work earns no points.

A response that provides a completely correct left Riemann sum and approximation $\int_0^{12} r'(t) dt$ (i.e., $3r'(0) + 4r'(3) + 3r'(7) + 2r'(10) = 3(-6.1) + 4(-5.0) + 3(-4.4) + 2(-3.8) = -59.1$) earns 1 of the 2 points.

A response that has any error in a left Riemann sum or evaluation for $\int_0^{12} r'(t) dt$ earns no points.

Units are not required or read in this part.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Form of right Riemann sum
- ☐ Answer

Solution:

$$\begin{aligned}
 \int_0^{12} r'(t) dt &\approx 3r'(3) + 4r'(7) + 3r'(10) + 2r'(12) \\
 &= 3(-5.0) + 4(-4.4) + 3(-3.8) + 2(-3.5) \\
 &= -51
 \end{aligned}$$

Part D

2-3-3

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first 2 points could be earned in either order.

A response with a completely correct product rule, missing one or both of the correct differentials, earns the product rule point, but not the chain rule point.

For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h + \frac{1}{3}\pi r^2$ earns the first point, but not the second.

A response that treats r or h (but not both) as a constant is eligible for the chain rule point but not the product rule point. For example, $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt}$ is correct if h is constant, and thus earns the chain rule point.

Note: Neither $\frac{dV}{dt} = \frac{2}{3}\pi r \frac{dh}{dt}$ nor $\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} \frac{dh}{dt}$ earns any points.

A response that assumes a functional relationship between r and h (such as $r = 2h$), and uses this relationship to create a function for volume in terms of one variable, is eligible for at most the chain rule point. For example,

$r = 2h \rightarrow V = \frac{1}{3}\pi(2h)^2 h = \frac{4}{3}\pi h^3 \rightarrow \frac{dV}{dt} = 4\pi h^2 \frac{dh}{dt}$ earns only the chain rule point.

A response that mishandles the constant $\frac{1}{3}\pi$ cannot earn the third point but is eligible for the first 2 points.

The third point cannot be earned without both of the first 2 points.

$\frac{dV}{dt} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2)$ earns all 3 points.

Units are not required or read in this part.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Product rule
- ☐ Chain rule
- ☐ Answer

Solution:

$$\frac{dV}{dt} = \frac{2}{3}\pi r h \frac{dr}{dt} + \frac{1}{3}\pi r^2 \frac{dh}{dt}$$

$$\left. \frac{dV}{dt} \right|_{t=3} = \frac{2}{3}\pi(100)(50)(-5) + \frac{1}{3}\pi(100)^2(-2) = -\frac{70,000\pi}{3}$$

The rate of change of the volume of the sculpture at $t = 3$ is approximately $-\frac{70,000\pi}{3}$ cubic centimeters per day.

2-3-3

47.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	2	4	6	8	10	12
$R(t)$ (vehicles per hour)	2935	3653	3442	3010	3604	1986	2201

On a certain weekday, the rate at which vehicles cross a bridge is modeled by the differentiable function R for $0 \leq t \leq 12$, where $R(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). Values of $R(t)$ for selected values of t are given in the table above.

Use the data in the table to approximate $R'(5)$. Show the computations that lead to your answer. Using correct units, explain the meaning of $R'(5)$ in the context of the problem.

Use a midpoint sum with three subintervals of equal length indicated by the data in the table to approximate the value of $\int_0^{12} R(t) \, dt$. Indicate units of measure.

On a certain weekend day, the rate at which vehicles cross the bridge is modeled by the function H defined by $H(t) = -t^3 - 3t^2 + 288t + 1300$ for $0 \leq t \leq 17$, where $H(t)$ is measured in vehicles per hour and t is the number of hours since 7:00 A.M. ($t = 0$). According to this model, what is the average number of vehicles crossing the bridge per hour on the weekend day for $0 \leq t \leq 12$?

For $12 < t < 17$, $L(t)$, the local linear approximation to the function H given in part (c) at $t = 12$, is a better model for the rate at which vehicles cross the bridge on the weekend day. Use $L(t)$ to find the time t , for $12 < t < 17$, at which the rate of vehicles crossing the bridge is 2000 vehicles per hour. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point does not require a simplified answer. Substitution of function values is required.

The second point requires a time reference, a rate, and units. The point may be earned with an incorrect approximation based on one computational error and provided the behavior described is consistent with the sign of the approximation, the numerical value, and the time reference.

2-3-3



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ approximation using values from table
- ☐ interpretation with units

Solution:

$$R'(5) \approx \frac{R(6) - R(4)}{6 - 4} = \frac{3010 - 3442}{2} = -216$$

At time $t = 5$ hours (12 p.m.), the rate at which vehicles cross the bridge is decreasing at a rate of approximately 216 vehicles per hour per hour.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

- The first point does not require a simplified answer. Substitution of function values is required. A single substitution error for one and only one of $R(2)$, $R(6)$, or $R(10)$ will still earn this point.

The second point requires earning the first point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ midpoint sum setup
- ☐ approximation using values from table with units

Solution:

$$\begin{aligned} \int_0^{12} R(t) \, dt &\approx 4(R(2) + R(6) + R(10)) \\ &= 4(3653 + 3010 + 1986) \\ &= 34,596 \text{ vehicles} \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with a correct integrand and limits of integration. The constant $\frac{1}{12}$ is not needed to earn this point.

2-3-3



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ definite integral
- ☐ answer

Solution:

$$\frac{1}{12-0} \int_0^{12} H(t) dt = 2452$$

On the weekend day for $0 \leq t \leq 12$, the average number of vehicles crossing the bridge is 2452 vehicles per hour.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ slope
- ☐ $L(t) = 2000$
- ☐ answer

Solution:

$$L(t) = H(12) + H'(12)(t - 12)$$

$$H(12) = 2596$$

$$H'(12) = -216$$

$$L(t) = 2000 \Rightarrow t = 14.759259$$

On the weekend day, the rate at which vehicles cross the bridge is 2000 vehicles per hour at time $t = 14.759$ hours.

2-3-3

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (minutes)	0	15	45	70	90	100
$A(t)$ (automobiles)	0	30	190	250	405	600

Prior to a sporting event, the number of automobiles that have entered a stadium parking lot is modeled by the differentiable function A , where t is the number of minutes since the parking lot opened. Values of $A(t)$ for selected values of t are given in the table above.

According to the model, what is the average rate at which automobiles enter the parking lot, in automobiles per minute, over the time interval $45 \leq t \leq 90$ minutes?

Write $A'(95)$ as the limit of a difference quotient. Use the data in the table to approximate $A'(95)$. Show the computations that lead to your answer.

What is the shortest time interval during which it is guaranteed that $A(t) = 400$ for some time t in the interval? Justify your answer.

For $0 \leq t \leq 45$, the function f defined by $f(t) = \frac{2}{27}t^2 + \frac{8}{9}t$ models the number of automobiles that have entered the parking lot, where t is the number of minutes since the parking lot opened. Find $f'(10)$, the rate at which automobiles enter the parking lot, in automobiles per minute, at time $t = 10$ minutes.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct average rate of change.

Solution:

$$\frac{A(90) - A(45)}{90 - 45} = \frac{405 - 190}{90 - 45} = \frac{215}{45} = \frac{43}{9}$$

2-3-3

The average rate at which automobiles enter the parking the time interval $45 \leq t \leq 90$ minutes is $\frac{43}{9}$ automobiles per minute.

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$A'(95) = \lim_{t \rightarrow 95} \frac{A(t) - A(95)}{t - 95}$$

$$\text{OR } A'(95) = \lim_{h \rightarrow 0} \frac{A(95+h) - A(95)}{h}$$

$$A'(95) \approx \frac{A(100) - A(90)}{100 - 90} = \frac{600 - 405}{100 - 90} = \frac{195}{10} = \frac{39}{2}$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ time interval
- ☐ justification, using Intermediate Value Theorem

Solution:

The shortest time interval for which it is guaranteed that $A(t) = 400$ is $70 \leq t \leq 90$.

2-3-3

Because A is differentiable, A is continuous.

$$A(70) = 250 < 400 < 405 = A(90)$$

By the Intermediate Value Theorem, there must be at least one time t , for $70 \leq t \leq 90$, such that $A(t) = 400$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $f'(t)$
- ☐ $f'(10)$

Solution:

$$f'(t) = \frac{4}{27}t + \frac{8}{9}$$

$$f'(10) = \frac{4}{27} \cdot 10 + \frac{8}{9} = \frac{40}{27} + \frac{24}{27} = \frac{64}{27}$$

At time $t = 10$ minutes, automobiles enter the parking lot at a rate of $\frac{64}{27}$ automobiles per minute.

2-3-3

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	1	3	6	8
$P(t)$ (passengers)	0	35	204	600	728

The number of passengers who have boarded a ship is modeled by the differentiable function P , where t is the number of hours since boarding began. Values of $P(t)$ for selected values of t are given in the table above.

- According to the model, what is the average rate at which passengers board the ship, in passengers per hour, over the time interval $1 \leq t \leq 8$ hours?
- Write $P'(4.5)$ as the limit of a difference quotient. Use the data in the table to approximate $P'(4.5)$. Show the computations that lead to your answer.
- Must there be a time t , for $3 \leq t \leq 6$, at which $P(t) = 500$? Justify your answer.
- The total number of gallons of water used by the passengers on the ship is modeled by the function G defined by $G(t) = 120t\sqrt{t}$ for $0 \leq t \leq 8$, where t is the number of hours since boarding began. Find $G'(4)$, the rate at which passengers use water, in gallons per hour, at time $t = 4$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct average rate of change.

Solution:

$$\frac{P(8) - P(1)}{8 - 1} = \frac{728 - 35}{7} = 99$$

The average rate at which passengers board the ship over the time interval $1 \leq t \leq 8$ hours is 99 passengers per hour.

Part B

2-3-3

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ limit expression
- ☐ approximation

Solution:

$$P'(4.5) = \lim_{t \rightarrow 4.5} \frac{P(t) - P(4.5)}{t - 4.5}$$

OR

$$P'(4.5) = \lim_{h \rightarrow 0} \frac{P(4.5+h) - P(4.5)}{h}$$

$$P'(4.5) \approx \frac{P(6) - P(3)}{6 - 3} = \frac{600 - 204}{3} = 132$$

Part C

1 out of 2 points if correct justification does not specifically reference IVT .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ P is continuous with reason
- ☐ justification, using Intermediate Value Theorem

Solution:

Because P is differentiable, P is continuous.

$$P(3) = 204 < 500 < 600 = P(6)$$

By the Intermediate Value Theorem, there must be at least one time t , for $3 \leq t \leq 6$, such that

2-3-3

$$P(t) = 500.$$

Part D

The first point is earned for demonstration of a correct derivative for G and does not require a separate statement of $G'(t)$.

The derivative and evaluation of the derivative can be expressed in one line to earn both points.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $G'(t)$

☐ $G'(4)$

Solution:

$$G(t) = 120t\sqrt{t} = 120t^{\frac{3}{2}}$$

$$G'(t) = 120 \cdot \frac{3}{2}t^{\frac{1}{2}} = 180\sqrt{t}$$

$$G'(4) = 180\sqrt{4} = 180 \cdot 2 = 360$$

At time $t = 4$ hours, passengers use water at a rate of 360 gallons per hour.

2-3-3

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(t) = \begin{cases} g(t) & \text{for } 0 \leq t \leq 12 \\ \frac{t^2}{8} - 3t + 83 & \text{for } 12 < t \leq 24 \end{cases}$$

t (hours)	0	4	6	8	12
$g(t)$ (degrees Fahrenheit)	87	81.5	76	70.5	65

The temperature of a room, in degrees Fahrenheit, on a certain day is modeled by the function f defined above, where g is a differentiable function and t is measured in hours. Values of $g(t)$ at selected values of t are given in the table above.

- According to the model f , what is the average rate of change of the temperature of the room over the time interval $0 \leq t \leq 12$ hours?
- Use the data in the table to approximate $f'(10)$, the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time $t = 10$ hours. Show the computations that lead to your answer.
- Is f continuous for $0 \leq t \leq 24$? Justify your answer.
- Find $f'(20)$, the instantaneous rate of change in the temperature of the room, in degrees Fahrenheit per hour, at time $t = 20$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct average rate of change.

Solution:

$$\frac{f(12) - f(0)}{12 - 0} = \frac{g(12) - g(0)}{12 - 0} = \frac{65 - 87}{12} = -\frac{11}{6}$$

2-3-3

The average rate of change of the temperature of the room over the time interval $0 \leq t \leq 12$ hours is $-\frac{11}{6}$ degrees Fahrenheit per hour

Part B

The second point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct approximation of $f'(10)$

Solution:

$$f'(10) \approx \frac{g(12) - g(8)}{12 - 8} = \frac{65 - 70.5}{4} = -\frac{5.5}{4}$$

Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that f is a polynomial on $12 < t \leq 24$ to explain why f is continuous on that interval to earn the second point. The third point is for continuity at $t = 12$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ differentiability of g implies continuity
- ☐ f is continuous on $12 < t \leq 24$
- ☐ $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$

Solution:

Because g is differentiable, g is continuous on $0 \leq t \leq 12$. $f(t) = g(t)$ for $0 \leq t \leq 12$.

Because f is a polynomial function on $12 < t \leq 24$, f is continuous on $12 < t \leq 24$.

$$\lim_{t \rightarrow 12} f(t) = f(12) = g(12) = 65$$

$$\lim_{t \rightarrow 12^+} f(t) = \frac{12^2}{8} - 3(12) + 83 = 65$$

2-3-3

Because $\lim_{t \rightarrow 12^-} f(t) = \lim_{t \rightarrow 12^+} f(t) = f(12)$, f is continuous at $t = 12$.

Thus, f is continuous on $0 \leq t \leq 24$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The student response accurately includes both of the criteria below.

- ☐ $f'(t)$
- ☐ $f'(20)$

Solution:

$$f(t) = \frac{t^2}{8} - 3t + 83$$

$$f'(t) = \frac{1}{4}t - 3$$

$$f'(20) = \frac{1}{4}(20) - 3 = 5 - 3 = 2$$

The instantaneous rate of change in the temperature of the room at $t = 20$ hours is 2 degrees Fahrenheit per hour.

2-3-3

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

$$f(t) = \begin{cases} 48t + t^2 - \frac{t^3}{12} & \text{for } 0 \leq t < 6 \\ g(t) & \text{for } 6 \leq t \leq 12 \end{cases}$$

t (hours)	6	8	10	12
$g(t)$ (cubic meters)	306	376	428	474

At an excavation site, the amount of dirt that has been removed, in cubic meters, is modeled by the function f defined above, where g is a differentiable function and t is measured in hours. Values of $g(t)$ at selected values of t are given in the table above.

- According to the model f , what is the average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours?
- Use the data in the table to approximate $f'(9)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 9$ hours. Show the computations that lead to your answer.
- Is f continuous for $0 \leq t \leq 12$? Justify your answer.
- Find $f'(2)$, the instantaneous rate of change in the amount of dirt removed, in cubic meters per hour, at time $t = 2$ hours.

Part A

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct average rate of change.

Solution:

$$\frac{f(12) - f(6)}{12 - 6} = \frac{g(12) - g(6)}{12 - 6} = \frac{474 - 306}{12 - 6} = \frac{168}{6} = 28$$

2-3-3

The average rate of change of the amount of dirt removed over the time interval $6 \leq t \leq 12$ hours is 28 cubic meters per hour.

Part B

The point is earned for a correct expression that substitutes both correct values from the table, with or without simplification.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The student response accurately includes a correct approximation of $f'(9)$.

Solution:

$$f'(9) \approx \frac{g(10) - g(8)}{10 - 8} = \frac{428 - 376}{10 - 8} = \frac{52}{2} = 26$$

Part C

The response must explicitly connect differentiability and continuity to earn the first point and use the fact that f is a polynomial for $0 \leq t < 6$ to explain why f is continuous on that interval to earn the second point. The third point is for continuity at $t = 6$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ f is continuous for $0 \leq t < 6$
- ☐ differentiability of g implies continuity
- ☐ $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$

Solution:

Because f is a polynomial function for $0 \leq t < 6$, f is continuous for $0 \leq t < 6$.

Because g is differentiable, g is continuous on $6 \leq t \leq 12$. $f(t) = g(t)$ for $6 \leq t \leq 12$.

$$\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^-} \left(48t + t^2 - \frac{t^3}{12} \right) = 48(6) + 6^2 - \frac{6^3}{12} = 288 + 36 - 18 = 306$$

$$\lim_{t \rightarrow 6^+} f(t) = f(6) = g(6) = 306$$

2-3-3

Because $\lim_{t \rightarrow 6^-} f(t) = \lim_{t \rightarrow 6^+} f(t) = f(6)$, f is continuous at $t = 6$.

Thus, f is continuous for $0 \leq t \leq 12$.

Part D

The first point is earned for demonstration of a correct derivative for f and does not require a separate statement of $f'(t)$, so $f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4}$ earns both points. The derivative and evaluation of the derivative can be expressed in one line to earn both points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ $f'(t)$

☐ $f'(2)$

Solution:

$$f(t) = 48t + t^2 - \frac{t^3}{12} \text{ for } 0 \leq t < 6$$

$$f'(t) = 48 + 2t - \frac{t^2}{4} \text{ for } 0 < t < 6$$

$$f'(2) = 48 + 2 \cdot 2 - \frac{2^2}{4} = 51$$

The instantaneous rate of change in the amount of dirt removed at $t = 2$ hours is 51 cubic meters per hour.

2-3-3

52. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

r (centimeters)	0	1	2	2.5	4
$f(r)$ (milligrams per square centimeter)	1	2	6	10	18

The density of a bacteria population in a circular petri dish at a distance r centimeters from the center of the dish is given by an increasing, differentiable function f , where $f(r)$ is measured in milligrams per square centimeter. Values of $f(r)$ for selected values of r are given in the table above.

(a) Use the data in the table to estimate $f'(2.25)$. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) The total mass, in milligrams, of bacteria in the petri dish is given by the integral expression $2\pi \int_0^4 r f(r) \, dr$.

Approximate the value of $2\pi \int_0^4 r f(r) \, dr$ using a right Riemann sum with the four subintervals indicated by the data in the table.

(c) Is the approximation found in part (b) an overestimate or underestimate of the total mass of bacteria in the petri dish? Explain your reasoning.

(d) The density of bacteria in the petri dish, for $1 \leq r \leq 4$, is modeled by the function g defined by $g(r) = 2 - 16(\cos(1.57\sqrt{r}))^3$. For what value of k , $1 < k < 4$, is $g(k)$ equal to the average value of $g(r)$ on the interval $1 \leq r \leq 4$?

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point the response must provide both a difference and a quotient and must explicitly use values of f from the table.

2-3-3

Simplification of the numerical value is not required to earn the first point. If the numerical value is simplified, it must be correct.

The interpretation requires all of the following: distance $r = 2.25$, density of bacteria (population) is increasing or changing, at a rate of 8, and units of milligrams per square centimeter per centimeter.

The second point (interpretation) cannot be earned without a nonzero presented value for $f'(2.25)$.

To earn the second point the interpretation must be consistent with the presented nonzero value for $f'(2.25)$. In particular, if the presented value for $f'(2.25)$ is negative, the interpretation must include “decreasing at a rate of $|f'(2.25)|$ ” or “changing at a rate of $f'(2.25)$.” The second point cannot be earned for an incorrect statement such as “the bacteria density is decreasing at a rate of $-8\dots$ ” even for a presented $f'(2.25) = -8$.

The units (mg/cm²/cm) may be attached to the estimate of $f'(2.25)$ and, if so, do not need to be repeated in the interpretation.

If units attached to the estimate do not agree with units in the interpretation, read the units in the interpretation.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Estimate
- ☐ Interpretation with units

Solution:

$$f'(2.25) \approx \frac{f(2.5) - f(2)}{2.5 - 2} = \frac{10 - 6}{0.5} = 8$$

At a distance of $r = 2.25$ centimeters from the center of the petri dish, the density of the bacteria population is increasing at a rate of 8 milligrams per square centimeter per centimeter.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The presence or absence of 2π has no bearing on earning the first point.

The first point is earned for a sum of four products with at most one error in any single value among the four products. Multiplication by 1 in any term does not need to be shown, but all other products must be explicitly shown.

A response of $1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)$ earns the first point but not the second.

A response with any error in the Riemann sum is not eligible for the second point.

2-3-3

A response that provides a completely correct left Riemann sum for $2\pi \int_0^4 r f(r) dr$ and approximation (91π) earns one of the two points. A response that has any error in a left Riemann sum or evaluation for $2\pi \int_0^4 r f(r) dr$ earns no points.

A response that provides a completely correct right Riemann sum for $2\pi \int_0^4 f(r) dr$ and approximation (80π) earns one of the two points. A response that has any error in a right Riemann sum or evaluation for $2\pi \int_0^4 f(r) dr$ earns no points.

Simplification of the numerical value is not required to earn the second point. If a numerical value is given, it must be correct to three decimal places.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation

Solution:

$$\begin{aligned}
 2\pi \int_0^4 r f(r) dr &\approx 2\pi(1 \cdot f(1) \cdot (1 - 0) + 2 \cdot f(2) \cdot (2 - 1) \\
 &\quad + 2.5 \cdot f(2.5) \cdot (2.5 - 2) + 4 \cdot f(4) \cdot (4 - 2.5)) \\
 &= 2\pi(1 \cdot 2 \cdot 1 + 2 \cdot 6 \cdot 1 + 2.5 \cdot 10 \cdot 0.5 + 4 \cdot 18 \cdot 1.5) \\
 &= 269\pi = 845.088
 \end{aligned}$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the second point a response must explain that $r f(r)$ is increasing and, therefore, the right Riemann sum is an overestimate. The second point can be earned without having earned the first point.

A response that attempts to explain based on a left Riemann sum for $2\pi \int_0^4 r f(r) dr$ from part (b) earns no points.

A response that attempts to explain based on a right Riemann sum for $2\pi \int_0^4 f(r) dr$ from part (b) earns no points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

2-3-3

- ☐ Product rule expression for $\frac{d}{dr}(rf(r))$
- ☐ Answer with explanation

Solution:

$$\frac{d}{dr}(rf(r)) = f(r) + rf'(r)$$

Because f is nonnegative and increasing, $\frac{d}{dr}(rf(r)) > 0$ on the interval $0 \leq r \leq 4$. Thus, the integrand $rf(r)$ is strictly increasing.

Therefore, the right Riemann sum approximation of $2\pi \int_0^4 rf(r) dr$ is an overestimate.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a definite integral, with or without $\frac{1}{4-1}$ or $\frac{1}{3}$.

A response that presents a definite integral with incorrect limits but a correct integrand earns the first point.

Presentation of the numerical value 9.875795 is not required to earn the second point. This point can be earned by the average value setup: $\frac{1}{3} \int_1^4 g(r) dr$.

Once earned for the average value setup, the second point cannot be lost. Subsequent errors will result in not earning the third point.

The third point is earned only for the value $k = 2.497$.

The third point cannot be earned without the second.

Special case: A response that does not provide the average value setup but presents an average value of -13.955 is using degree mode on their calculator. This response would not earn the second point but could earn the third point for an answer of $k = 2.5$ (or 2.499).



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Definite integral
- ☐ Average value
- ☐ Answer

Solution:

2-3-3

$$\text{Average value} = g_{\text{avg}} = \frac{1}{4-1} \int_1^4 g(r) \, dr$$

$$\frac{1}{4-1} \int_1^4 g(r) \, dr = 9.875795$$

$$g(k) = g_{\text{avg}} \Rightarrow k = 2.497$$

2-3-3

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{3}$	$\frac{2}{3}$	1
$R(t)$ (gallons per hour)	11	8	5	0

The rate at which water leaks from a container is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 1$. Values of $R(t)$ are given in the table above for selected values of t .

- (a) Use the data in the table to find an approximation for $R'(\frac{1}{2})$. Show the computations that lead to your answer. Indicate units of measure.
- (b) Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate $\int_0^1 R(t) \, dt$. Indicate units of measure.
- (c) Use the data in the table to evaluate $\int_0^{\frac{1}{3}} R'(t) \, dt$.
- (d) The sum $\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \frac{1}{2n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

☐ approximation

2-3-3

☐ units
Solution:

$$R' \left(\frac{1}{2} \right) \approx \frac{R \left(\frac{2}{3} \right) - R \left(\frac{1}{3} \right)}{\frac{2}{3} - \frac{1}{3}} = \frac{5 - 8}{\frac{1}{3}} = -9 \text{ gallons per hour per hour}$$

Part B

To earn the first point, the response must present the form of a left Riemann sum. Therefore, $\frac{1}{3}(11 + 8 + 5)$ earns the first and second points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ left Riemann sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{3} \left(R(0) + R\left(\frac{1}{3}\right) + R\left(\frac{2}{3}\right) \right) \\ &= \frac{1}{3}(11 + 8 + 5) = 8 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -3 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

$$\int_0^{\frac{1}{3}} R'(t) \, dt = R\left(\frac{1}{3}\right) - R(0) = 8 - 11 = -3$$

2-3-3

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length $\frac{1}{2}$. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

✓

0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{2n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{2n}$. Thus, the integral is over an interval of length $\frac{1}{2}$. This interval can be taken to be $\left[\frac{1}{4}, \frac{3}{4}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{4} + \frac{k}{2n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{4} + \frac{k}{2n}\right) \cdot \frac{1}{2n} \right) = \int_{\frac{1}{4}}^{\frac{3}{4}} R(t) \, dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+\frac{1}{2}} R\left(\frac{1}{4} - a + t\right) dt.$

2-3-3

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (hours)	0	$\frac{1}{2}$	1	$\frac{3}{2}$	2
$R(t)$ (gallons per hour)	16	12	7	3	0

The rate at which oil leaks from a drum is modeled by the twice-differentiable function R , where $R(t)$ is measured in gallons per hour and t is measured in hours for $0 \leq t \leq 2$. Values of $R(t)$ are given in the table above for selected values of t .

(a) Use the data in the table to find an approximation for $R'\left(\frac{3}{4}\right)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Use a trapezoidal sum with the four subintervals indicated by the data in the table to approximate $\int_0^2 R(t) \, dt$. Indicate units of measure.

(c) Use the data in the table to evaluate $\int_0^1 R'(t) \, dt$.

(d) The sum $\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \frac{1}{n}$ is a right Riemann sum with n subintervals of equal length. The limit of this sum as n goes to infinity can be interpreted as a definite integral. Express the limit as a definite integral.

Part A

Numerical answers do not need to be simplified for the approximation point. The approximation must include a substitution of the function values from the table into the difference quotient.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ approximation
units

2-3-3

**Solution:**

$$R' \left(\frac{3}{4} \right) \approx \frac{R(1) - R\left(\frac{1}{2}\right)}{1 - \frac{1}{2}} = \frac{7 - 12}{\frac{1}{2}} = -10$$

gallons per hour per hour

Part B

To earn the first point, the response must present the form of a trapezoidal sum. Therefore,

$$\frac{1}{2} \left(\frac{1}{2}(16 + 12) + \frac{1}{2}(12 + 7) + \frac{1}{2}(7 + 3) + \frac{1}{2}(3 + 0) \right) \text{ earns the first and second points.}$$

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ trapezoidal sum
- ☐ approximation
- ☐ units

Solution:

$$\begin{aligned} \int_0^1 R(t) \, dt &\approx \frac{1}{2} \left(\frac{1}{2} (R(0) + R(\frac{1}{2})) + \frac{1}{2} (R(\frac{1}{2}) + R(1)) + \frac{1}{2} (R(1) + R(\frac{3}{2})) + \frac{1}{2} (R(\frac{3}{2}) + R(2)) \right) \\ &= \frac{1}{2} \left(\frac{1}{2} (16 + 12) + \frac{1}{2} (12 + 7) + \frac{1}{2} (7 + 3) + \frac{1}{2} (3 + 0) \right) \\ &= \frac{1}{4} (28 + 19 + 10 + 3) = 15 \text{ gallons} \end{aligned}$$

Part C

The response should include supporting work for the numerical answer of -9 .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct value.

Solution:

2-3-3

$$\int_0^1 R'(t) dt = R(1) - R(0) = 7 - 16 = -9$$

Part D

The first point may be earned as presented in the solution or by using that information to determine that the interval of integration is of length 1. A response that produces the correct definite integral earns all 3 points.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

✓

0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ $\Delta t = \frac{1}{n}$
- ☐ limits of integration
- ☐ integrand

Solution:

For this Riemann sum, $\Delta t = \frac{1}{n}$. Thus, the integral is over an interval of length 1. This interval can be taken to be $\left[\frac{1}{2}, \frac{3}{2}\right]$.

The right endpoints of the subintervals used in the Riemann sum would be of the form $t_k = \frac{1}{2} + \frac{k}{n}$ for k from 1 to n .

Therefore, $\lim_{n \rightarrow \infty} \left(\sum_{k=1}^n R\left(\frac{1}{2} + \frac{k}{n}\right) \cdot \frac{1}{n} \right) = \int_{\frac{1}{2}}^{\frac{3}{2}} R(t) dt.$

Note: The limit of the Riemann sum can be written as any definite integral of the form $\int_a^{a+1} R\left(\frac{1}{2} - a + t\right) dt.$

2-3-3

55.  A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

t (years)	0	10	22	30	40
$A(t)$ (hundred pounds per year)	163	142	124	118	113

The amount of nitrogen in a particular river is affected by two sources. Nitrogen from the first source enters the river at a rate modeled by the decreasing, differentiable function A . Selected values of $A(t)$ are shown in the table above. Nitrogen from the second source enters the river at a rate modeled by the function B defined by

$B(t) = 20 + \left(\frac{95}{1+\sqrt[3]{t}}\right)^{1.25}$. Both $A(t)$ and $B(t)$ are measured in hundreds of pounds of nitrogen per year, and t is measured in years for $0 \leq t \leq 40$.

(a) Using a right Riemann sum with the four subintervals indicated by the data in the table, approximate the total amount of nitrogen from the first source that enters the river over the time interval $0 \leq t \leq 40$. Is this approximation an overestimate or an underestimate? Give a reason for your answer.

(b) Find $B'(39)$. Using correct units, interpret the meaning of $B'(39)$ in the context of the problem.

(c) Based on model B , find the average rate, in hundreds of pounds per year, at which nitrogen from the second source enters the river during the time interval $5 \leq t \leq 35$.

(d) Using model B and your approximation from part (a), find an approximation for the total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$. Give your answer to the nearest hundred pounds.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for a sum of four products with at most one error among the 8 factors. If there is an error in the setup, the second point is not earned.

2-3-3

A response of $10 \cdot A(10) + 12 \cdot A(22) + 8 \cdot A(30) + 10 \cdot A(40)$ would earn the first point, but not yet the second. The first two points are earned for $10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113$.

Without the products, the sum $1420 + 1488 + 944 + 1130$ by itself would earn the second point but not the first point.

Once the first point is earned it cannot be lost by incorrectly using values from the table or performing some type of operation on the Riemann sum.

A completely correct setup and approximation using a left Riemann sum $(10 \cdot 163 + 12 \cdot 142 + 8 \cdot 124 + 10 \cdot 118 = 5506)$ will earn one of the first two points and is not eligible for the third point. Any error in the setup or evaluation using a left Riemann sum will earn no points.

Units are not scored in this part, except an approximation of 498,200 must clearly be labeled as *pounds* to earn the second point.

We will interpret the word “function” to mean A , unless otherwise prescribed.

The response must indicate which function is decreasing. A response of “underestimate because it (or the graph, or the curve) is decreasing” will not earn the third point.

The third point will not be earned if the response concludes the approximation is an underestimate because the table values are decreasing.

The third point will not be earned if incorrect statements are made in addition to saying A is decreasing.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Right Riemann sum setup
- ☐ Approximation
- ☐ Underestimate with reason

Solution:

$$\begin{aligned} \int_0^{40} A(t) dt &\approx (10 - 0) \cdot A(10) + (22 - 10) \cdot A(22) \\ &\quad + (30 - 22) \cdot A(30) + (40 - 30) \cdot A(40) \\ &= 10 \cdot 142 + 12 \cdot 124 + 8 \cdot 118 + 10 \cdot 113 = 4982 \end{aligned}$$

Approximately 4982 hundred pounds of nitrogen from the first source enters the river over the time interval $0 \leq t \leq 40$.

The approximation is an underestimate because a right Riemann sum is used and A is decreasing.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

2-3-3

The value of $B'(39)$ may appear in the interpretation. No work is required in order to earn the first point.

In the presence of a correct value for $B'(39)$, any symbolic attempt at differentiation will be ignored.

Any unevaluated expression equivalent to $1.25 \left(\frac{95}{1 + \sqrt[3]{39}} \right)^{0.25} \left(\frac{-95}{(1 + \sqrt[3]{39})^2} \right) \left(\frac{1}{3}(39)^{-2/3} \right)$ will earn the first point.

The second point cannot be earned if no value for $B'(39)$ is presented.

The units (hundred pounds per year per year) may be attached to the value of $B'(39)$ and if so, they do not need to be repeated in the interpretation.

If units attached to $B'(39)$ do not agree with units in the interpretation, read the units in the interpretation.

The interpretation must include time $t = 39$, rate of change of the rate at which nitrogen enters the river (or that the rate that nitrogen enters the river is decreasing at the stated rate), and units of hundred pounds per year per year.

The second point cannot be earned with an incorrect statement such as “... is decreasing at a rate of -0.385 ...” or the statement “... is increasing at a rate of -0.385 ...”

Use of the term “acceleration” of the amount of nitrogen coming from the second source will not earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Value
- ☐ Interpretation

Solution:

$$B'(39) = -0.384963$$

At time $t = 39$ years, the rate at which nitrogen from the second source enters the river was decreasing at a rate of 0.385 (or 0.384) hundred pounds per year per year.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for presenting the definite integral $\int_5^{35} B(t) dt$.

Dividing by 30 will be scored in the second point.

Once the first point is earned, it cannot be lost.

If the first point is not earned because of a copy error in the integrand, the response can earn the second point for the correct answer.

2-3-3

An unsupported answer of 80.063 does not earn the second point.

Units are not scored in this part, except an answer of 8006.3 must clearly be labeled as *pounds per year* in order to earn the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Definite integral
- ☐ Answer

Solution:

$$\frac{1}{35-5} \int_5^{35} B(t) dt = 80.063385$$

During the time interval $5 \leq t \leq 35$, the average rate at which nitrogen from the second source enters the river was 80.063 hundred pounds per year.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

For any $c > 0$, the presence of $c \int_0^{40} B(t) dt$ will earn the first point.

A copy error in the integrand imported from part (c) will earn the first point. Any new copy error will not earn the first point. In either case the second point can be earned if the correct answer is given.

Read for correctness or consistency with any positive value of $\int_0^{40} A(t) dt$ imported from part (a).

The second point is not earned for a correct value with no supporting work.

The second point is earned for an answer that is one of the four values 8368, 8369, 8368.35, or 8368.351 (with supporting work).

If one of the values 8368, 8369, 8368.35, or 8368.351 is presented but then rounded to 8400 (e.g., $8368 \approx 8400$), the response will earn the second point. However, a response of just 8400, without one of these four values present, will not earn the second point.

Units are not scored in this part, except an answer of 836,800 (or 836,900) must clearly be labeled as *pounds* in order to earn the second point.



0	1	2
---	---	---

2-3-3

The student response accurately includes **both** of the criteria below.

☐ $\int_0^{40} B(t) \, dt$

☐ Answer

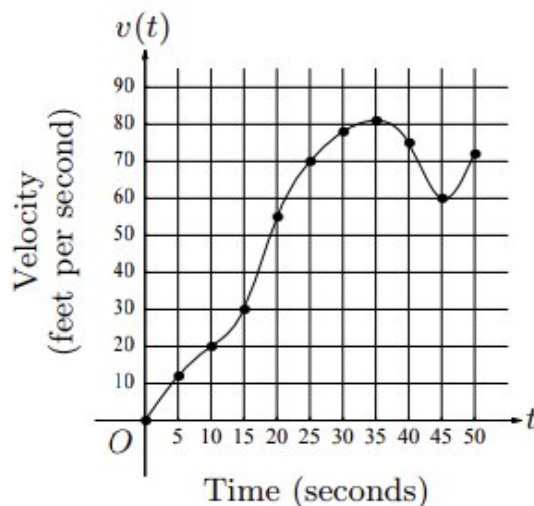
Solution:

$$\text{Total} = \int_0^{40} A(t) \, dt + \int_0^{40} B(t) \, dt \approx 4982 + 3386.350701 \approx 8368$$

The total combined amount of nitrogen from the two sources that enters the river over the time interval $0 \leq t \leq 40$ is approximately 8368 hundred pounds.

2-3-3

56.



t (seconds)	$v(t)$ (feet per second)
0	0
5	12
10	20
15	30
20	55
25	70
30	78
35	81
40	75
45	60
50	72

The graph of the velocity $v(t)$, in ft/sec, of a car traveling on a straight road, for $0 \leq t \leq 50$, is shown above. A table of values for $v(t)$, at 5 second intervals of time t , is shown to the right of the graph.

- (a) During what intervals of time is the acceleration of the car positive? Give a reason for your answer
- (b) Find the average acceleration of the car, in ft/sec^2 , over the interval $0 \leq t \leq 50$.
- (c) Find one approximation for the acceleration of the car, in ft/sec^2 , at $t = 40$. Show the computations you used to arrive at your answer.
- (d) Approximate $\int_0^{50} v(t) dt$ with a Riemann sum, using the midpoints of five subintervals of equal length. Using correct units, explain the meaning of this integral.

Part A

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$



0	1	2	3
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The student response earns three of the following points:

1 point is earned for (0, 35)

1 point is earned for (45, 50)

1 point is earned for the reason

Acceleration is positive on (0, 35) and (45, 50) because the velocity $v(t)$ is increasing on $[0, 35]$ and $[45, 50]$

2-3-3

Part B

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

$$\text{Avg. Acc.} = \frac{v(50) - v(0)}{50 - 0} = \frac{72 - 0}{50} = \frac{72}{50}$$

or 1.44 ft/sec^2

Part C

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

2-3-3



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the method

1 point is earned for the answer

Difference quotient; e.g

$$\frac{v(45) - v(40)}{5} = \frac{60 - 75}{5} = -3 \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{5} = \frac{75 - 81}{5} = -\frac{6}{5} \text{ ft/sec}^2 \text{ or}$$

$$\frac{v(40) - v(35)}{10} = \frac{60 - 81}{10} = -\frac{21}{10} \text{ ft/sec}^2$$

Part D

1 point is earned for midpoint Riemann sum

1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} & \int_0^{50} v(t) dt \\ & \approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ & = 10(12 + 30 + 70 + 81 + 60) \\ & = 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.



0	1	2	3
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The student response earns all of the following points:

1 point is earned for midpoint Riemann sum

2-3-3

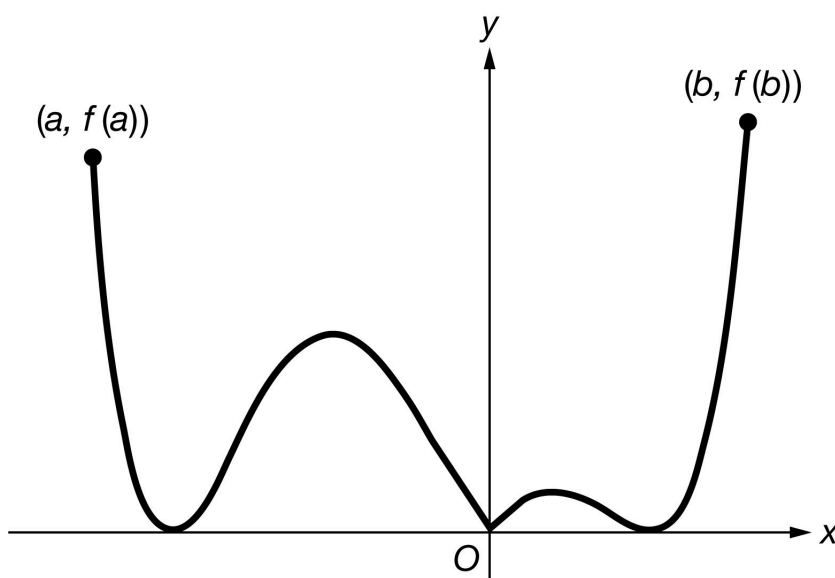
1 point is earned for answer

1 point is earned for meaning of integral

$$\begin{aligned} \int_0^{50} v(t) dt \\ &\approx 10 [v(5) + v(15) + v(25) + v(35) + v(45)] \\ &= 10(12 + 30 + 70 + 81 + 60) \\ &= 2530 \text{ feet} \end{aligned}$$

This integral is the total distance traveled in feet over the time 0 to 50 seconds.

57.



The graph of a function f with $f(b) > f(a)$ is shown above for $a \leq x \leq b$. The derivative of f exists for all x in the interval $a < x < b$ except $x = 0$. For how many values of c , for $a < c < b$, does $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \frac{f(b) - f(a)}{b - a}$?

- (A) Zero
- (B) Two
- (C) Three
- (D) Four



Answer D

Correct. The expression $\frac{f(b) - f(a)}{b - a}$ is the average rate of change of f on the interval $a \leq x \leq b$ and is equal to the slope of the secant line passing through the points $(a, f(a))$ and $(b, f(b))$. The expression $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ is the instantaneous rate of change of f at $x = c$ and is equal to the slope of the line tangent to the graph of f at $x = c$. There are four points on the graph of f at which the slope of the tangent line is equal to the slope of the secant line. Two of the points occur slightly to the left of the interior local maxima. Two of the points occur slightly to the right of the local minima with horizontal tangents.

2-3-3

58.



x	0	1
$f(x)$	1	2

Let f be the function given by $f(x) = 2^{x^2}$. Selected values of f are given in the table above. If the values in the table are used to approximate $f'(0.5)$, what is the difference between the approximation and the actual value of $f'(0.5)$?

(A) 0

(B) 0.176

(C) 0.824

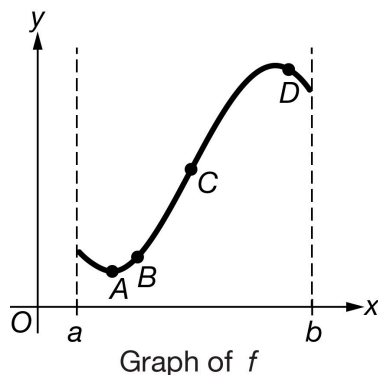
(D) 1

**Answer B**

Correct. The numerical value of the derivative at $x = 0.5$ obtained from the calculator is $f'(0.5) = 0.824$. A difference quotient can be used with the values in the table to estimate the derivative as $\frac{f(1)-f(0)}{1-0} = \frac{2-1}{1} = 1$. The error between the actual derivative value and this approximation is 0.176.

2-3-3

59.



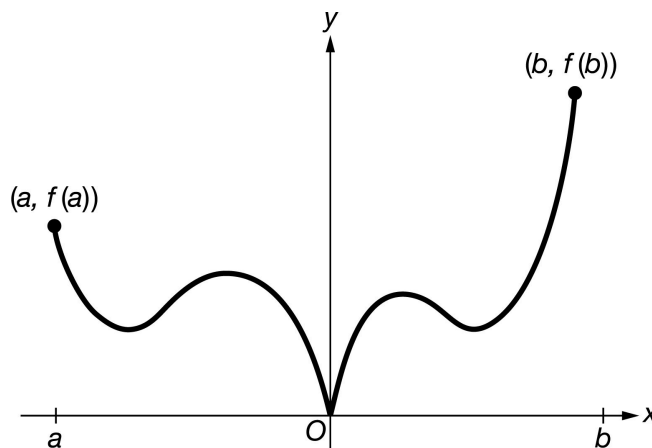
The graph of the trigonometric function f is shown above for $a \leq x \leq b$. At which of the following points on the graph of f could the instantaneous rate of change of f equal the average rate of change of f on the interval $[a, b]$?

(A) A (B) B (C) C (D) D **Answer B**

Correct. The instantaneous rate of change of f at the point B is the slope of the line tangent to the graph of f at the point B . The average rate of change of f on the interval $[a, b]$ is the slope of the secant line through the points $(a, f(a))$ and $(b, f(b))$. The tangent line at B appears to be parallel to the secant line. Therefore, the instantaneous rate of change at B could be equal to the average rate of change.

2-3-3

60.

Graph of f

The graph of a function f with $f(b) > f(a)$ is shown above for $a \leq x \leq b$. The derivative of f exists for all x in the interval $a < x < b$ except $x = 0$. For how many values of c , for $a < c < b$, does $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \frac{f(b) - f(a)}{b - a}$?

- (A) Zero
- (B) Two
- (C) Three
- (D) Four

**Answer D**

Correct. The expression $\frac{f(b) - f(a)}{b - a}$ is the average rate of change of f over the interval $a \leq x \leq b$ and is equal to the slope of the secant line passing through the points $(a, f(a))$ and $(b, f(b))$. The expression $\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$ is the instantaneous rate of change of f at $x = c$ and is equal to the slope of the line tangent to the graph of f at $x = c$. There are four points on the graph of f at which the slope of the tangent line is equal to the slope of the secant line. Two of the points occur to the left of the y -axis, and two of the points occur to the right of the y -axis.

2-3-3

61.



x	0	1
$f(x)$	1	2

Let f be the function given by $f(x) = 2^{x^3}$. Selected values of f are given in the table above. If the values in the table are used to approximate $f'(0.5)$, what is the difference between the approximation and the actual value of $f'(0.5)$?

(A) 0

(B) 0.433

(C) 0.567

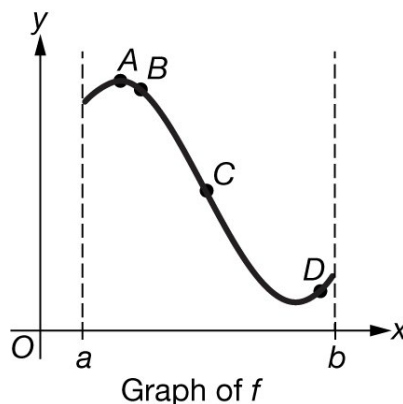
(D) 1

**Answer B**

Correct. The numerical value of the derivative at $x = 0.5$ obtained from the calculator is $f'(0.5) = 0.567$. A difference quotient can be used with the values in the table to estimate the derivative as $\frac{f(1)-f(0)}{1-0} = \frac{2-1}{1} = 1$. The error between the actual derivative value and this approximation is 0.433.

2-3-3

62.



The graph of the trigonometric function f is shown above for $a \leq x \leq b$. At which of the following points on the graph of f could the instantaneous rate of change of f equal the average rate of change of f on the interval $[a, b]$?

- (A) A
 (B) B
 (C) C
 (D) D



Answer B

Correct. The instantaneous rate of change of f at the point B is the slope of the line tangent to the graph of f at the point B . The average rate of change of f on the interval $[a, b]$ is the slope of the secant line through the points $(a, f(a))$ and $(b, f(b))$. The tangent line at B appears to be parallel to the secant line. Therefore, the instantaneous rate of change at B could be equal to the average rate of change.

63.  Let f be the function given by $f(x) = \cos x - \csc x$. What is the value of $f'(1)$?

- (A) $f'(1)$ is undefined.
 (B) -0.648
 (C) -0.078
 (D) 0



Answer C

Correct. $f'(1)$ is the notation for the derivative of f at $x = 1$. This is the instantaneous rate of change of f at $x = 1$. The numerical value of the derivative at $x = 1$ obtained from the graphing calculator is $f'(1) = -0.078$.

2-3-3

64.  Let f be the function given by $f(x) = \frac{1}{7}x^7 + \frac{1}{2}x^6 - x^5 - \frac{15}{4}x^4 + \frac{4}{3}x^3 + 6x^2$. Which of the following statements is true?
- (A) $f'(-3.1) < f'(-1.5) < f'(0.4)$
- (B) $f'(-3.1) < f'(0.4) < f'(-1.5)$
- (C) $f'(-1.5) < f'(0.4) < f'(-3.1)$
- (D) $f'(0.4) < f'(-1.5) < f'(-3.1)$ ✓

Answer D

Correct. The calculator is used to store the expression for $f(x)$ and to find the numerical derivative values at the three values of x . They are, in increasing order, $f'(0.4) = 4.387$, $f'(-1.5) = 4.922$, and $f'(-3.1) = 14.974$.

65.  Let f be the function given by $f(x) = e^{-x} + \cos x - 1$. What is the value of $f'(2)$?
- (A) -1.281
- (B) -1.140
- (C) -1.045 ✓
- (D) -1

Answer C

Correct. $f'(2)$ is the notation for the derivative of f at $x = 2$. This is the instantaneous rate of change of f at $x = 2$. The numerical value of the derivative at $x = 2$ obtained from the graphing calculator is $f'(2) = -1.045$.

66.  Let f be the function given by $f(x) = x^4 + x^3 - 3x^2 - x - \cos(x^2)$. Of the following values of x , at which does the line tangent to the graph of f have the greatest slope?
- (A) $x = -2$
- (B) $x = -1$ ✓
- (C) $x = 0$
- (D) $x = 1$

Answer B

Correct. The slope of the line tangent to the graph of f at a point is the value of the derivative f' at that point. The calculator is used to store the expression for $f(x)$ and to find the numerical derivative values at the four values of x . They are, in increasing order, $f'(-2) = -5.973$, $f'(0) = -1$, $f'(1) = 1.683$, and $f'(-1) = 2.317$.

2-3-3

67.  Let f be the function given by $f(x) = \frac{1}{7}x^7 - \frac{7}{6}x^6 + 3x^5 - \frac{5}{4}x^4 - \frac{16}{3}x^3 + 6x^2$. Which of the following statements is true?
- (A) $f'(-1.1) < f'(0.5) < f'(1.4)$
- (B) $f'(-1.1) < f'(1.4) < f'(0.5)$
- (C) $f'(0.5) < f'(1.4) < f'(-1.1)$
- (D) $f'(1.4) < f'(0.5) < f'(-1.1)$ ✓

Answer D

Correct. The calculator is used to store the expression for $f(x)$ and to find the numerical derivative values at the three values of x . They are, in increasing order, $f'(1.4) = -0.774$, $f'(0.5) = 2.109$, and $f'(-1.1) = 9.102$.

68.  Let f be the function given by $f(x) = x^4 + \frac{1}{2}x^3 - 5x^2 + \tan\left(\frac{x}{2}\right)$. Of the following values of x , at which does the line tangent to the graph of f have the greatest slope?
- (A) $x = -2$
- (B) $x = -1$ ✓
- (C) $x = 0$
- (D) $x = 1$

Answer B

Correct. The slope of the line tangent to the graph of f at a point is the value of the derivative f' at that point. The calculator is used to store the expression for $f(x)$ and to find the numerical derivative values at the four values of x . They are, in increasing order, $f'(-2) = -4.287$, $f'(1) = -3.851$, $f'(0) = 0.500$, and $f'(-1) = 8.149$.

2-3-3

69.  On a certain workday, the rate, in tons per hour, at which unprocessed gravel arrives at a gravel processing plant is modeled by $G(t) = 90 + 45\cos\left(\frac{t^2}{18}\right)$, where t is measured in hours and $0 \leq t \leq 8$. At the beginning of the workday ($t = 0$), the plant has 500 tons of unprocessed gravel. During the hours of operation, $0 \leq t \leq 8$, the plant processes gravel at a constant rate of 100 tons per hour.
- (a) Find $G'(5)$. Using correct units, interpret your answer in the context of the problem.
- (b) Find the total amount of unprocessed gravel that arrives at the plant during the hours of operation on this workday.
- (c) Is the amount of unprocessed gravel at the plant increasing or decreasing at time $t = 5$ hours? Show the work that leads to your answer.
- (d) What is the maximum amount of unprocessed gravel at the plant during the hours of operation on this workday? Justify your answer.

Part A

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $G'(5)$

1 point is earned for the interpretation with units

$$G'(5) = -24.588 \text{ (or } -24.587)$$

The rate at which gravel is arriving is decreasing by 24.588 (or 24.587) tons per hour per hour at time $t = 5$ hours.

Part B

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$



0	1	2
---	---	---

2-3-3

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\int_0^8 G(t) dt = 825.551 \text{ tons}$$

Part C

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for comparing $G(5)$ to 100

1 point is earned for the conclusion

$$G(5) = 98.140764 < 100$$

At time $t = 5$, the rate at which unprocessed gravel is arriving is less than the rate at which it is being processed.

Therefore, the amount of unprocessed gravel at the plant is decreasing at time $t = 5$.

Part D

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

2-3-3

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $A'(t) = 0$

1 point is earned for the answer

1 point is earned for the justification

The amount of unprocessed gravel at time t is given by $A(t) = 500 + \int_0^t (G(s) - 100)ds$.

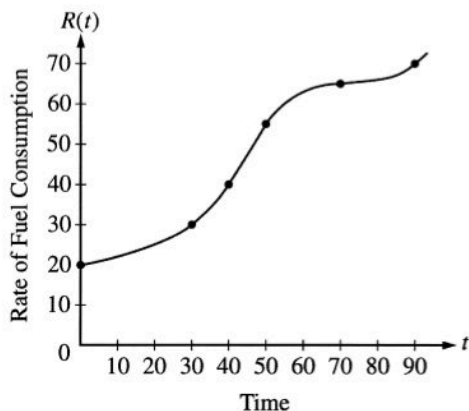
$$A'(t) = G(t) - 100 = 0 \Rightarrow t = 4.923480$$

t	$A(t)$
0	500
4.92348	635.376123
8	525.551089

The maximum amount of unprocessed gravel at the plant during this workday is 635.376 tons.

2-3-3

70.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown above.

(a) Use data from the table to find an approximation for $R'(45)$. Show the computations that lead to your answer. Indicate units of measure.

(b) The rate of fuel consumption is increasing fastest at time $t = 45$ minutes. What is the value of $R''(45)$? Explain your reasoning.

(c) Approximate the value of $\int_0^{90} R(t) dt$ using a left Riemann sum with the five subintervals indicated by the data in the table. Is this numerical approximation less than the value of $\int_0^{90} R(t) dt$? Explain your reasoning.

(d) For $0 < b \leq 90$ minutes, explain the meaning of $\int_0^b R(t) dt$ in terms of fuel consumption for the plane. Explain the meaning of $\frac{1}{b} \int_0^b R(t) dt$ in terms of fuel consumption for the plane. Indicate units of measure in both answers.

Part A

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the appropriate difference quotient using numbers from table and interval that contains 45

2-3-3

$$R'(45) \approx \frac{R(50) - R(40)}{50 - 40} = \frac{55 - 40}{10} \\ = 1.5 \text{ gal/min}^2$$

1 point is earned for the correct answer of 1.5 gal/min^2 with correct units

Part B

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer of $R''(45) = 0$

1 point is earned for the reason that $R''(45) = 0$

$R''(45) = 0$ since $R'(t)$ has a maximum at $t = 45$.

Part C

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct value of left Riemann sum

$$\int_0^{90} R(t) dt \approx (30)(20) + (10)(30) + (10)(40) + (20)(55) + (20)(65) = 3700$$

2-3-3

1 point is earned for answer "less" with reason

Yes, this approximation is less because the graph of R is increasing on the interval.

Part D

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the meaning of $\int_0^b R(t)dt$

$\int_0^b R(t)dt$ is the total amount of fuel in gallons consumed for the first b minutes.

<-1> if no reference to time b

1 point is earned for the meaning of $\frac{1}{b} \int_0^b R(t)dt$

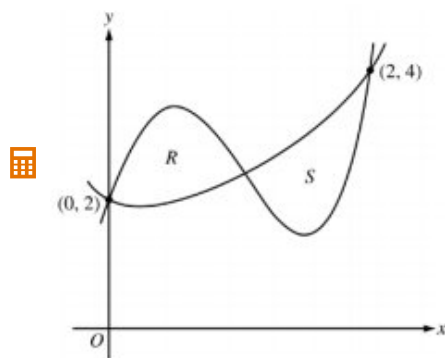
$\frac{1}{b} \int_0^b R(t)dt$ is the average value of the rate of fuel consumption in gallons/min during the first b minutes.

<-1> if no reference to time b

1 point is earned for the correct units in both answers

2-3-3

71.



Let f and g be the functions defined by $f(x) = 1 + x + e^{x^2-2x}$ and $g(x) = x^4 - 6.5x^2 + 6x + 2$. Let R and S be the two regions enclosed by the graphs of f and g shown in the figure above.

(a) Find the sum of the areas of regions R and S .

(b) Region S is the base of a solid whose cross sections perpendicular to the x -axis are squares. Find the volume of the solid.

(c) Let h be the vertical distance between the graphs of f and g in region S . Find the rate at which h changes with respect to x when $x = 1.8$.

Part A

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

$$\begin{aligned} \text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

2 points are earned for the integrands

1 point is earned for the answer

The graphs of $y = f(x)$ and $y = g(x)$ intersect in the first quadrant at the points $(0, 2)$, $(2, 4)$, and

$(A, B) = (1.032832, 2.401108)$.

2-3-3

$$\begin{aligned}\text{Area} &= \int_0^A [g(x) - f(x)]dx + \int_A^2 [f(x) - g(x)]dx \\ &= 0.997427 + 1.006919 = 2.004\end{aligned}$$

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_A^2 [f(x) - g(x)]^2 dx = 1.283$$

Part C

1 point is earned for considering h'

1 point is earned for the answer

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for considering h'

1 point is earned for the answer

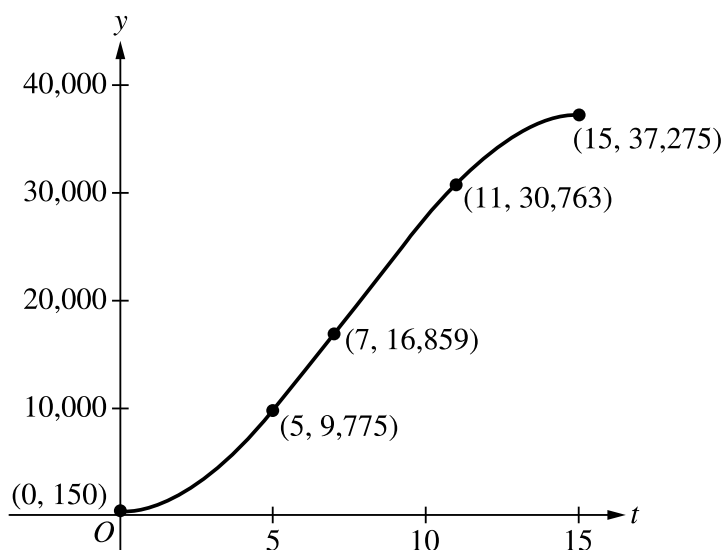
2-3-3

$$h(x) = f(x) - g(x)$$

$$h'(x) = f'(x) - g'(x)$$

$$h'(1.8) = f'(1.8) - g'(1.8) = -3.812 \text{ (or } -3.811)$$

72.



An airplane's height above sea level is modeled by the differentiable function $y = h(t)$, where t is the number of minutes after takeoff and $h(t)$ is measured in feet. The graph of h is shown in the figure for $0 \leq t \leq 15$. Of the following, which is the best estimate of the rate at which the plane is rising above sea level, in feet per minute, at time $t = 9$?

- (A) $\frac{37,275-150}{15-0}$
 (B) $\frac{37,275-9,775}{15-5}$
 (C) $\frac{37,275-16,859}{15-7}$
 (D) $\frac{30,763-16,859}{11-7}$

**Answer D**

Correct. The symmetric difference quotient on the interval $[7, 11]$ is used to approximate the instantaneous rate of change at $t = 9$.

$$\frac{h(11)-h(7)}{11-7} = \frac{30,763-16,859}{11-7}$$

The average rate of change on the interval $[7, 11]$ is the slope of the secant line between the points at $t = 7$ and $t = 11$. The shape of the graph suggests that this secant line is almost parallel to the line tangent to the graph at time $t = 9$. Therefore, the slope of this secant line is a good approximation to the slope of the tangent line.

2-3-3

73.

t (seconds)	0	10	20	30	40	50	60	70	80
$v(t)$ (feet per second)	5	14	22	29	35	40	44	47	49

Rocket A has positive velocity $v(t)$ after being launched upward from an initial height of 0 feet at time $t = 0$ seconds. The velocity of the rocket is recorded for selected values of t over the interval $0 \leq t \leq 80$ seconds, as show in the table above.

(a) Find the average acceleration of rocket A over the time interval $0 \leq t \leq 80$ seconds. Indicate units of measure.

(b) Using correct units, explain the meaning of $\int_{10}^{70} v(t) dt$ in terms of the rockets flight. Use a midpoint Riemann sum with 3 subintervals of equal length to approximate $\int_{10}^{70} v(t) dt$.

(c) Rocket B is launched upward with an acceleration of $a(t) = \frac{3}{\sqrt{t+1}}$ feet per second per second. At time $t = 0$ seconds, the initial height of the rocket is 0 feet, and the initial velocity is 2 feet per second. Which of the two rockets is traveling faster at time $t = 80$ seconds? Explain your answer.

Part A

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$



0	1
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The student response earns all of the following points:

1 point is earned for the answer

Average acceleration of rocket A is

$$\frac{v(80) - v(0)}{80 - 0} = \frac{49 - 5}{80} = \frac{11}{20} \text{ ft/sec}^2$$

Part B

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

2-3-3

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the explanation

1 point is earned if uses $v(20)$, $v(40)$, $v(60)$

1 point is earned for the value

Since the velocity is positive, $\int_{10}^{70} v(t) dt$ represents the distance, in feet, traveled by rocket A from $t = 10$ seconds to $t = 70$ seconds.

A midpoint Riemann sum is

$$20[v(20) + v(40) + v(60)]$$

$$= 20[22 + 35 + 44] = 2020 \text{ ft}$$

Part C

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let $v_B(t)$ be the velocity of rocket B at time t .

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

2-3-3



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for $6\sqrt{t+1}$

1 point is earned for constant of integration

1 point is earned for uses initial condition

1 point is earned for $v_B(80)$, compares to $v(80)$, and draws a conclusion

Let $v_B(t)$ be the velocity of rocket B at time t .

$$v_B(t) = \int \frac{3}{\sqrt{t+1}} dt = 6\sqrt{t+1} + C$$

$$2 = v_B(0) = 6 + C$$

$$v_B(t) = 6\sqrt{t+1} - 4$$

$$v_B(80) = 50 > 49 = v(80)$$

Rocket B is traveling faster at time $t = 80$ seconds.

2-3-3

74.

x	2	3	5	8	13
$f(x)$	1	4	-2	3	6

Let f be a function that is twice differentiable for all real numbers. The table above gives values of f for selected points in the closed interval $2 \leq x \leq 13$.

(a) Estimate $f'(4)$. Show the work that leads to your answer.

(b) Evaluate $\int_2^{13} (3 - 5f'(x)) dx$. Show the work that leads to your answer.

(c) Use a left Riemann sum with subintervals indicated by the data in the table to approximate $\int_2^{13} f(x) dx$. Show the work that leads to your answer.

(d) Suppose $f'(5) = 3$ and $f''(x) < 0$ for all x in the closed interval $5 \leq x \leq 8$. Use the line tangent to the graph of f at $x = 5$ to show that $f(7) \leq 4$. Use the secant line for the graph of f on $5 \leq x \leq 8$ to show that $f(7) \geq 4/3$.

Part A

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$



0

1

The student response earns all of the following points:

1 point is earned for the answer

$$f'(4) \approx \frac{f(5) - f(3)}{5 - 3} = -3$$

Part B

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned} \int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3 dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8 \end{aligned}$$

2-3-3



0	1	2
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The student response earns all of the following points:

1 point is earned if uses Fundamental Theorem of Calculus

1 point is earned for the answer

$$\begin{aligned}\int_2^{13} (3 - 5f'(x)) dx &= \int_2^{13} 3dx - 5 \int_2^{13} f'(x) dx \\ &= 3(13 - 2) - 5(f(13) - f(2)) = 8\end{aligned}$$

Part C

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$



0	1	2
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The student response earns all of the following points:

1 point is earned for left Riemann sum

1 point is earned for the answer

$$\int_2^{13} f(x) dx \approx f(2)(3 - 2) + f(3)(5 - 3) + f(5)(8 - 5) + f(8)(13 - 8) = 18$$

Part D

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

2-3-3

An equation for the tangent line is $y = -2 + 3(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for showing $f(7) \leq 4$

1 point is earned for secant line

1 point is earned if shows $f(7) \geq 4/3$

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Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the line tangent to the graph of $y = f(x)$ at $x = 5$ lies above the graph for all x in the interval $5 < x \leq 8$.

Therefore, $f(7) \leq -2 + 3 \cdot 2 = 4$.

An equation for the secant line is $y = -2 + \frac{5}{3}(x - 5)$.

Since $f''(x) < 0$ for all x in the interval $5 \leq x \leq 8$, the secant line connecting $(5, f(5))$ and $(8, f(8))$ lies below the graph of $y = f(x)$ for all x in the interval $5 < x < 8$.

Therefore, $f(7) \geq -2 + \frac{5}{3} \cdot 2 = \frac{4}{3}$.

2-3-3

75.



t (days)	$W(t)$ (°C)
0	20
3	31
6	28
9	24
12	22
15	21

The temperature, in degrees Celsius (°C), of the water in a pond is a differentiable function W of time t . The table above shows the water temperature as recorded every 3 days over a 15-day period.

(a) Use data from the table to find an approximation for $W'(12)$. Show the computations that lead to your answer. Indicate units of measure.

(b) Approximate the average temperature, in degrees Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using a trapezoidal approximation with subintervals of length $\Delta t = 3$ days.

(c) A student proposes the function P , given by $P(t) = 20 + 10te^{(-t/3)}$, as a model for the temperature of the water in the pond at time t , where t is measured in days and $P(t)$ is measured in degrees Celsius. Find $P'(12)$. Using appropriate units, explain the meaning of your answer in terms of water temperature.

(d) Use the function P defined in part (c) to find the average value, in degrees Celsius, of $P(t)$ over the time interval $0 \leq t \leq 15$ days.

Part A

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the difference quotient

1 point is earned for the answer (with units)

Difference quotient; e.g.

2-3-3

$$W'(12) \approx \frac{W(15) - W(12)}{15 - 12} = -\frac{1}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(12) - W(9)}{12 - 9} = -\frac{2}{3}^{\circ}\text{C/day or}$$

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = -\frac{1}{2}^{\circ}\text{C/day}$$

Part B

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the trapezoidal method

1 point is earned for the answer

$$\frac{3}{2}(20 + 2(31) + 2(28) + 2(24) + 2(22) + 21) = 376.5$$

$$\text{Average temperature} \approx \frac{1}{15}(376.5) = 25.1^{\circ}\text{C}$$

Part C

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^{\circ}\text{C/day} \end{aligned}$$

This means the the temperature is decreasing at the rate of $0.549^{\circ}\text{C/day}$ when $t = 12$ days.

2-3-3



0	1	2
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The student response earns all of the following points:

1 point is earned for $P'(12)$ (with or without units)

1 point is earned for the interpretation

$$\begin{aligned} P'(12) &= 10e^{-t/3} - \frac{10}{3}te^{-t/3} \Big|_{t=12} \\ &= -30e^{-4} = -0.549^\circ\text{C/day} \end{aligned}$$

This means the the temperature is decreasing at the rate of 0.549°C/day when $t = 12$ days.

Part D

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ\text{C}$$



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits and average value constant

1 point is earned for the answer

$$\frac{1}{15} \int_0^{15} (20 + 10te^{-t/3}) dt = 25.757^\circ\text{C}$$

2-3-3

76.

t (minutes)	0	4	9	15	20
$W(t)$ (degrees Fahrenheit)	55.0	57.1	61.8	67.9	71.0

 The temperature of water in a tub at time t is modeled by a strictly increasing, twice-differentiable function W , where $W(t)$ is measured in degrees Fahrenheit and t is measured in minutes. At time $t = 0$, the temperature of the water is 55°F . The water is heated for 30 minutes, beginning at time $t = 0$. Values of $W(t)$ at selected times t for the first 20 minutes are given in the table above.

(a) Use the data in the table to estimate $W'(12)$. Show the computations that lead to your answer. Using correct units, interpret the meaning of your answer in the context of this problem.

(b) Use the data in the table to evaluate $\int_0^{20} W'(t) dt$. Using correct units, interpret the meaning of $\int_0^{20} W'(t) dt$ in the context of this problem.

(c) For $0 \leq t \leq 20$, the average temperature of the water in the tub is $\frac{1}{20} \int_0^{20} W(t) dt$. Use a left Riemann sum with the four subintervals indicated by the data in the table to approximate $\frac{1}{20} \int_0^{20} W(t) dt$. Does this approximation overestimate or underestimate the average temperature of the water over these 20 minutes? Explain your reasoning.

(d) For $20 \leq t \leq 25$, the function W that models the water temperature has first derivative given by $W'(t) = 0.4\sqrt{t} \cos(0.06t)$. Based on the model, what is the temperature of the water at time $t = 25$?

Part A

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the estimation

1 point is earned for the interpretation with units

$$W'(12) \approx \frac{W(15) - W(9)}{15 - 9} = \frac{67.9 - 61.8}{6} \\ = 1.017 \text{ (or } 1.016)$$

2-3-3

The water temperature is increasing at a rate of approximately 1.017°F per minute at time $t = 12$ minutes.

Part B

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.



0	1	2
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The student response earns all of the following points:

1 point is earned for the value

$$\int_0^{20} W'(t) dt = W(20) - W(0) = 71.0 - 55.0 = 16$$

1 point is earned for the interpretation with units

The water has warmed by 16°F over the interval from $t = 0$ to $t = 20$ minutes.

Part C

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned} \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\ &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\ &= \frac{1}{20} \cdot 1215.8 = 60.79 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

2-3-3



0	1	2	3
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The student response earns all of the following points:

1 point is earned for the left Riemann sum

1 points are earned the approximation

1 point is earned for the underestimate with reason

$$\begin{aligned}
 \frac{1}{20} \int_0^{20} W(t) dt &\approx \frac{1}{20} (4 \cdot W(0) + 5 \cdot W(4) + 6 \cdot W(9) + 5 \cdot W(15)) \\
 &= \frac{1}{20} (4 \cdot 55.0 + 5 \cdot 57.1 + 6 \cdot 61.8 + 5 \cdot 67.9) \\
 &= \frac{1}{20} \cdot 1215.8 = 60.79
 \end{aligned}$$

This approximation is an underestimate, because a left Riemann sum is used and the function W is strictly increasing.

Part D

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$



0	1	2
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The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned}
 W(25) &= 71.0 + \int_{20}^{25} W'(t) dt \\
 &= 71.0 + 2.043155 = 73.043
 \end{aligned}$$